MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

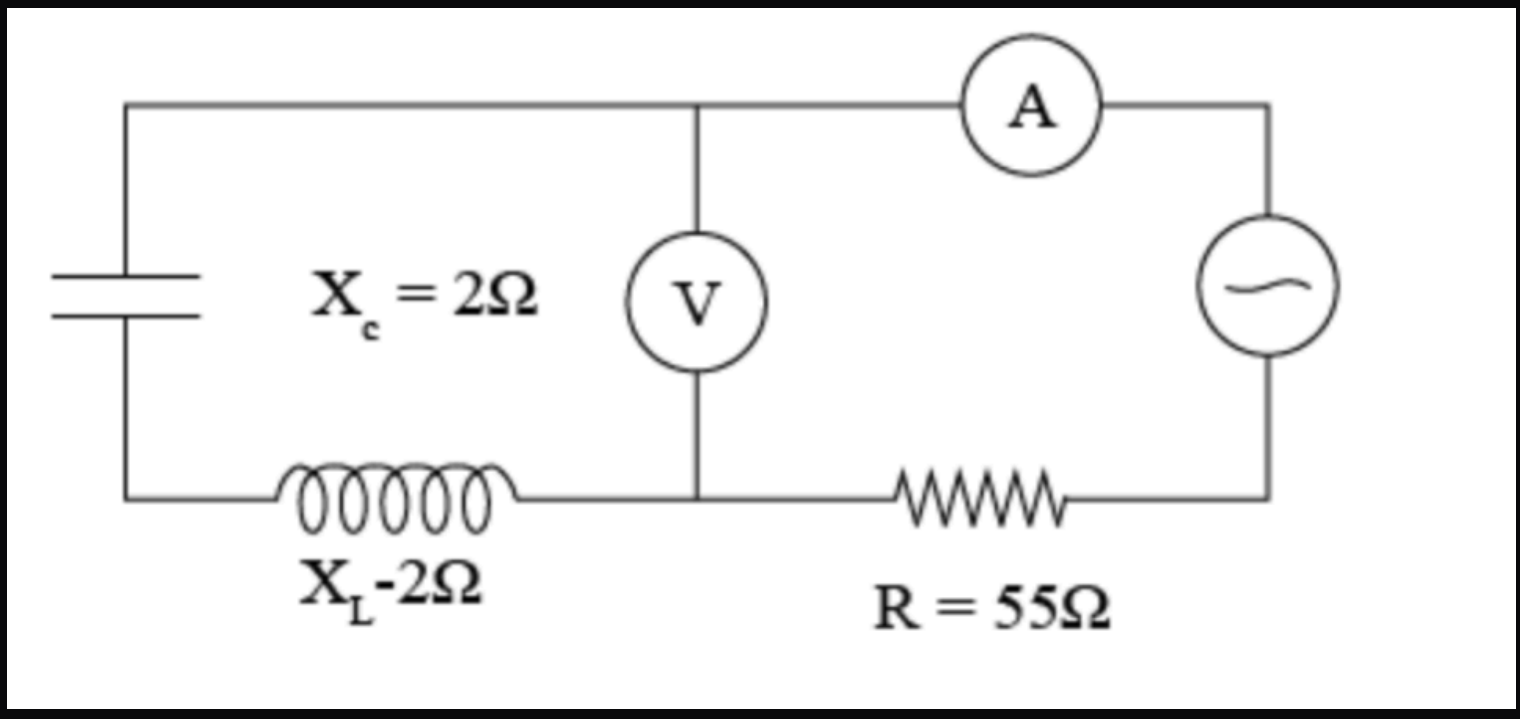
GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

The reading of the ammeter and voltmeter are (Both the instruments are ac meters and measures rms value)(take $V = 110\ V$)



(A) $2\ A, 110\ V$

(B) $2\ A, 0\ V$

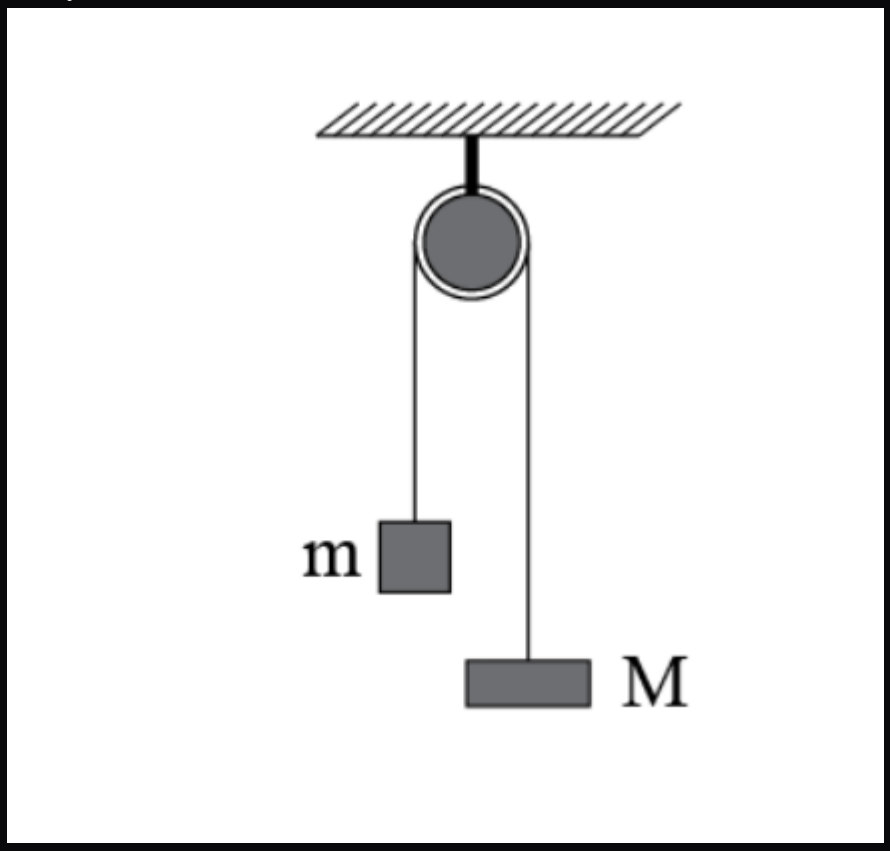
(C) $2\ A, 55\ V$

(D) $1\ A, 0\ V$

Q.2

Two masses m and $M(> m)$ are joined by a light string passing over a smooth light pulley. The centre of mass of

the system moves with an acceleration



(A) $g\left(\frac{M-m}{M+m}\right)$ downward

(B) $g\left(\frac{M-m}{M+m}\right)^2$ downward

(C) $g\left(\frac{M-m}{M+m}\right)^2$ upward if $\mathrm{M} < \mathrm{m}$

(D) zero

Q.3

Two non-ideal batteries are connected in series. Consider the following statements:
(A) The equivalent emf is larger than either of the two EMFs irrespective even if battery is reversed.
(B) The equivalent internal resistance is smaller than either of the two internal resistances.

(A) Each of A and B is correct.

(B) A is correct but B is wrong.

(C) B is correct but A is wrong.

(D) Each of A and B is wrong.

Q.4

If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Young's modulus will be:

(A) $\left[\mathrm{V}^{-2}\mathrm{A}^{2}\mathrm{F}^{2}\right]$

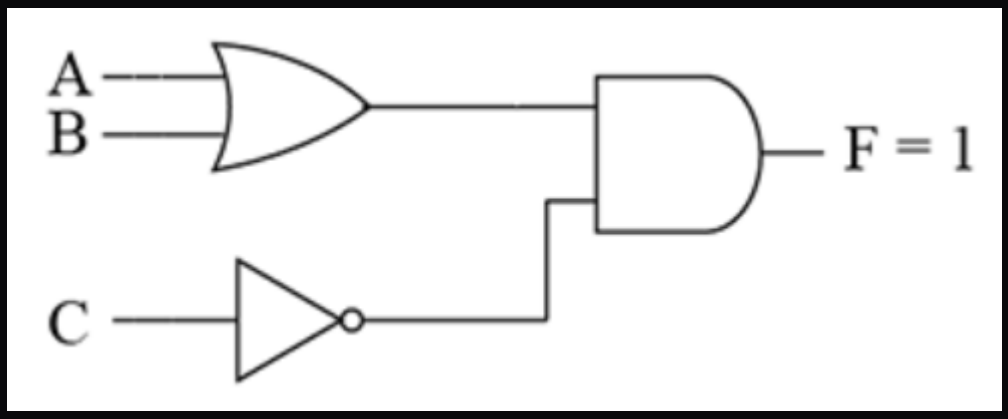
(B) $\left[\mathrm{V}^{-2}\mathrm{A}^{2}\mathrm{F}^{-2}\right]$

(C) $\left[\mathrm{V}^{-4}\mathrm{A}^{-2}\mathrm{F}\right]$

(D) $\left[\mathrm{V}^{-4} \mathrm{A}^{2} \mathrm{F}\right]$

Q.5

For given logic diagram, output $F = 1$, then inputs are



(A) $A = 0, B = 0, C = 0$

(B) $A = 0, B = 1, C = 0$

(C) $A = 1, B = 1, C = 1$

(D) $A = 0, B = 0, C = 1$

Q.6

A balloon of volume $1500\ \mathrm{m}^3$ and weighing $1650\ \mathrm{kg}$ with all its equipment is filled with He (density $0.2\ \mathrm{kg\ m}^{-3}$). If the density of air be $1.3\ \mathrm{kgm}^{-3}$, the pull on the rope tied to the balloon will be

(A) $300\ \mathrm{kg}$

(B) $1950\ \mathrm{kg}$

(C) $1650\ \mathrm{kg}$

(D) zero

Q.7

The minimum and maximum distance of a satellite from the centre of the earth are $2\mathrm{R}$ and $4\mathrm{R}$, respectively, where R is the radius of earth and M is the mass of earth. The radius of curvature at the point of maximum distance is

(A) $\frac{8R}{3}$

(B) $\frac{4\mathrm{R}}{3}$

(C) $\frac{3R}{8}$

(D) $\frac{3R}{4}$

Q.8

Assertion: Two concentric charged spherical shells are given. The potential difference between the shells depends on charge of inner shell.

Reason: Potential due to charge of outer shell remains same at every point inside the sphere.

(A) Assertion is true, reason is true. Reason is a correct explanation for assertion.

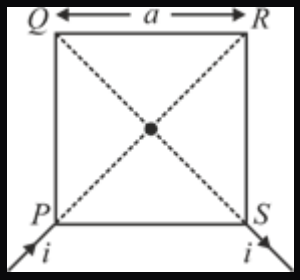
(B) Assertion is true, reason is true. Reason is not a correct explanation for assertion.

(C) Assertion is true, reason is false.

(D) Assertion is false but reason is true.

Q.9

In a square loop $PQRS$ made with a wire of cross-section current i enters from point P and leaves from point S. The magnitude of magnetic field induction at the centre O of the square is



(A) $\frac{\mu_0}{4\pi}\frac{2\sqrt{2}i}{a}$

(B) $\frac{\mu_0}{4\pi}\frac{4\sqrt{2}i}{a}$

(C) $\frac{\mu_0}{4}\frac{2\sqrt{2}i}{a}$

(D) zero

Q.10

The value of binding energy per nucleon of $^{40}_{20}\text{Ca}$ nucleus is

Given:

Mass of $^{40}_{20}\text{Ca}$ nucleus $= 39.962589\,u$

Mass of proton $= 1.007825\,u$

Mass of neutron $= 1.008665\,u$

and $1\,u = 931\ \text{MeV}\ \text{C}^{-2}$

(A) $18.32\ \text{MeV}$

(B) $8.55\ \text{MeV}$

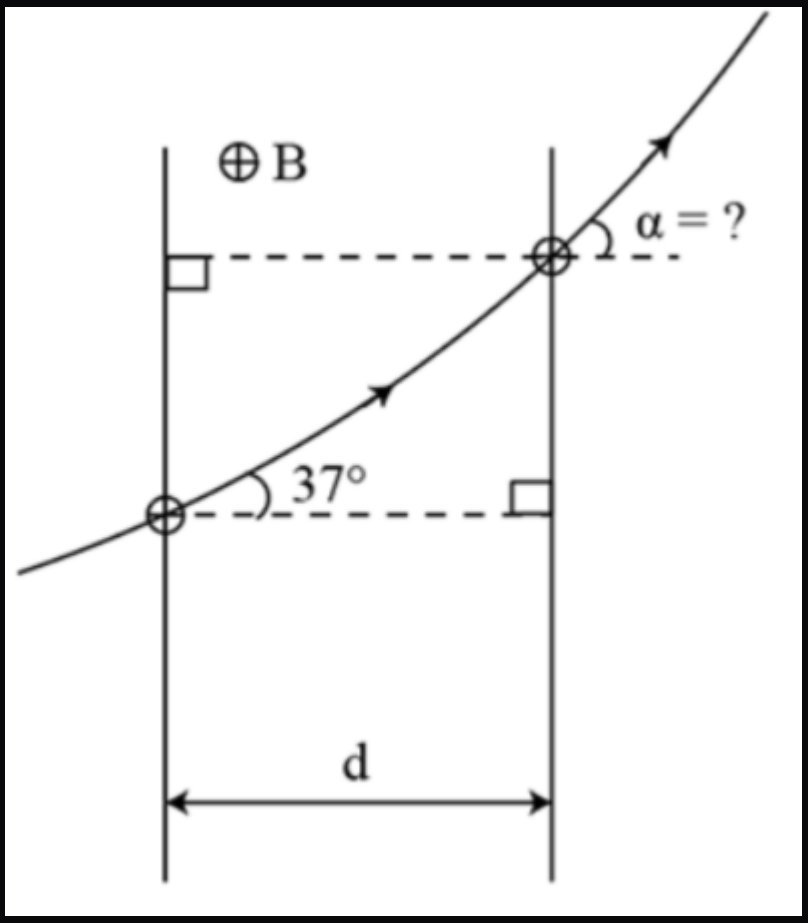
(C) $9.94\ \text{MeV}$

(D) $14.72\ \text{MeV}$

Q.11

A proton (mass m) accelerated by a potential difference V flies through a uniform transverse magnetic field B. The field occupies a region of space by width d'. If initial angle of proton is 37°, then final angle of proton α is :

$\left(d=\frac{\sqrt{2mV}}{5\sqrt{q}B}, \sin 37^\circ \approx \frac{3}{5}\right)$



(A) $\alpha = 45^\circ$

(B) $\alpha = 60^\circ$

(C) $\alpha = 75^\circ$

(D) $\alpha = \sin^{-1}\left(\frac{4}{5}\right) \approx 53^\circ$

Q.12

Match the forces given in column I with the properties given in column II :

Column I	Column II
i. Electric force of nucleus on electron	a. Conservative
ii. Your weight, i.e., the force that earth exerts on you	b. Action reaction force
iii. Force between earth and sun	c. Depends on the nature of medium between the interacting objects
iv. Repulsive force between two protons	d. Principle of superposition applies provided that other force, which are acting on the object, are also of this nature

(A) i. $\rightarrow$ a.,b.,c. ii. $\rightarrow$ a.,b. iii. $\rightarrow$ a.,b. iv. $\rightarrow$ a.,b.,c.

(B) i. $\rightarrow$ a.,b.,d. ii. $\rightarrow$ a.,b. iii. $\rightarrow$ a.,b.,d. iv. $\rightarrow$ a.,b.,c.

(C) i. $\rightarrow$ a.,d. ii. $\rightarrow$ a.,c. iii. $\rightarrow$ b.,c. iv. $\rightarrow$ b.,d.

(D) i. $\rightarrow$ a.,c. ii. $\rightarrow$ b.,c.,d. iii. $\rightarrow$ b.,c. iv. $\rightarrow$ a.,b.,d.

Q.13

A faulty thermometer has ice point at $-5^{\circ}\mathrm{C}$ and steam point at $105^{\circ}\mathrm{C}$. What would be the temperature shown by this thermometer of human body? (Assume correct temperature of human body is $37^{\circ}\mathrm{C}$)

(A) $34.8^{\circ}\mathrm{C}$

(B) $39.3^{\circ}\mathrm{C}$

(C) $35.7^{\circ}\mathrm{C}$

(D) $41.5^{\circ}\mathrm{C}$

Q.14

Statement - 1: Two tuning forks having frequency 410 Hz and 524 Hz are kept close and made to vibrate. Beats will not be heard.

Statement - 2: Sound waves superimpose only when the frequencies of superposing waves are equal or nearly equal.

(A) Statement - 1 is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement - 1.

(B) Statement - 1 is True, Statement - 2 is True; Statement - 2 is NOT a correct explanation for Statement - 1.

(C) Statement - 1 is True, Statement - 2 is False.

(D) Statement - 1 is False, Statement - 2 is True.

Q.15

Calculate the ratio of rms speeds of oxygen gas molecules to that of hydrogen gas molecules kept at the same temperature.

(A) 1:4

(B) 1:8

(C) 1:2

(D) 1:6

Q.16

Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $x_P(t) = at + bt^2$ and $x_Q(t) = ft - t^2$. At what time do the cars have the same velocity

(A) $\frac{f-a}{2(1+b)}$

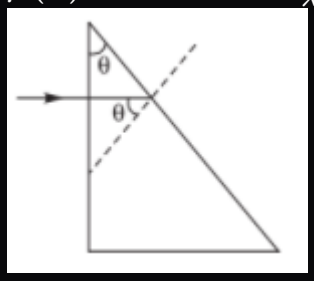
(B) $\frac{a-1}{1+b}$

(C) $\frac{a+1}{2(b-1)}$

(D) $\frac{a+f}{2(1+b)}$

Q.17

Parallel beam containing light of $\lambda = 400\ \text{nm}$ and $500\ \text{nm}$ is incident on a prism as shown in figure. The refractive index μ of the prism is given by the relation,
$\mu(\lambda) = 1.20 + \frac{0.8\times10^{-14}}{\lambda^2}$ (given $\sin\theta$ is slightly greater than 0.8)



Which of the following statement is correct?

(A) Light of $\lambda = 400\ \text{nm}$ undergoes total internal reflection

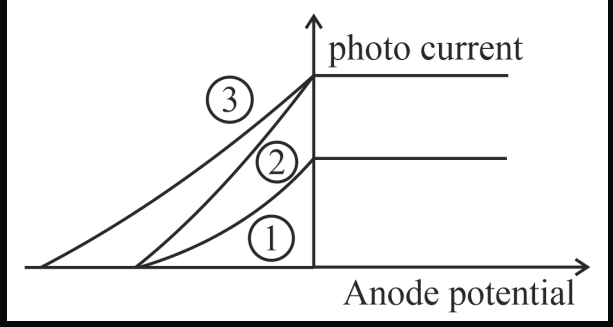
(B) Light of $\lambda = 500\ \text{nm}$ undergoes total internal reflection

(C) Neither of the two wavelengths undergoes total internal reflection

(D) Both wavelengths undergoes total internal reflection

Q.18

The following graph represents the variation of photo current with anode potential for a metal surface. Here $I_1,\ I_2$ and I_3 represents intensities and $\gamma_1, \gamma_2, \gamma_3$ represents frequency for curves $1,\ 2$ and 3 respectively, then



(A) $\gamma_1 = \gamma_2$ and $I_1 = I_2$

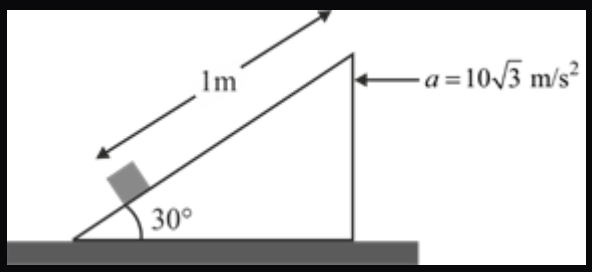
(B) $\gamma_2 = \gamma_3$ and $I_1 = I_3$

(C) $\gamma_1 = \gamma_2$ and $I_1 \neq I_2$

(D) $\gamma_1 = \gamma_3$ and $I_1 = I_3$

Q.19

In the figure, the wedge is pushed with an acceleration of $10\sqrt{3}\ \mathrm{m\ s^{-2}}$. It is seen that the block starts climbing up on the smooth inclined face of wedge. What will be the time taken by the block to reach the top?



(A) $\frac{2}{\sqrt{5}}$ s

(B) $\frac{1}{\sqrt{5}}$ s

(C) $\sqrt{5}$ s

(D) $\frac{\sqrt{5}}{2}$ s

Q.20

In Young's double-slit experiment, the distance between the two identical slits is 6.1 times larger than the slit width. Then the number of intensity maxima observed within the central maximum of the single slit diffraction pattern is :

(A) 3

(B) 6

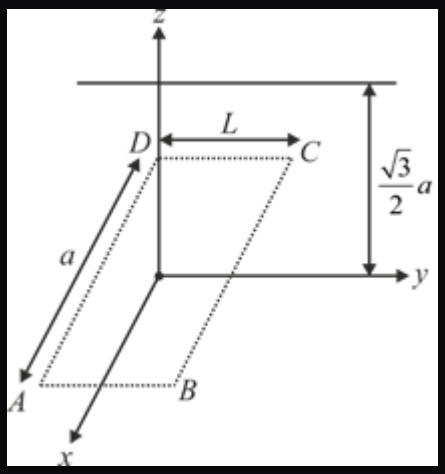
(C) 24

(D) 12

Q.21

An infinitely long uniform line charge distribution of charge per unit length λlies parallel to the y-axis in the $y - z$ plane at $z = \frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric field through the rectangular surface A B

C D lying in the $x - y$ plane with its center at the origin is $\frac{\lambda L}{n\varepsilon_0}$ (ε_0 = permittivity of free space), then the value of n is



Q.22

The primary and secondary coils of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in webers, t is time in second and ϕ_0 is a constant, the output voltage across the secondary coil is equal to 60 k V, then k is

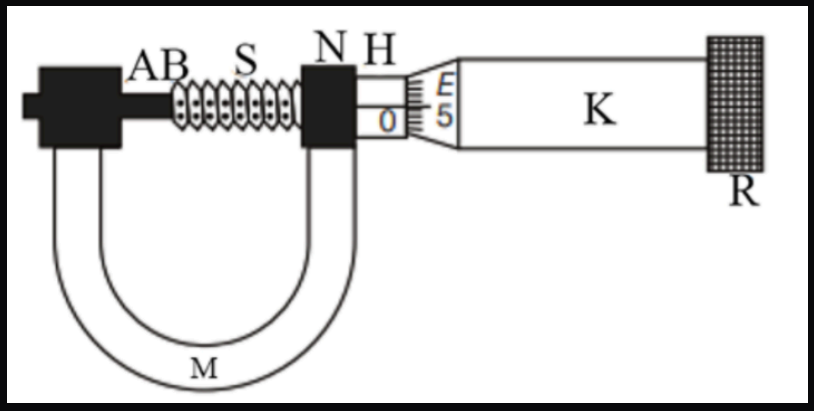
Q.23

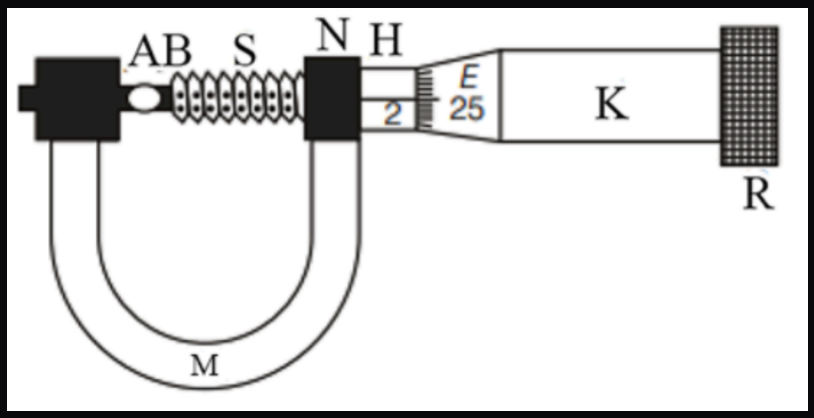
Two waves executing simple harmonic motion travelling in the same direction with same amplitude and frequency are superimposed. The resultant amplitude is equal to the $\sqrt{3}$ times of amplitude of individual motions. The phase difference between the two motions is (degree)

Q.24

The circular scale of a screw gauge has 50 divisions and pitch of 0.5 mm.

If main scale reading is 2, then the diameter of sphere is . . . . . micrometer.



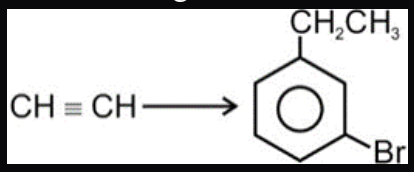


Q.25

An EM waves enters from medium-(1) to medium-(2). The electric fields are $\overrightarrow{E}_1 = E_{01}\sin(kx - \omega t)\hat{j}$ in medium-(1) \& $\overrightarrow{E}_2 = E_{02}\sin(3kx - \omega t)\hat{j}$ in medium-(2).If $\frac{\mu_{r_1}}{\mu_{r_2}} = \frac{1}{4}$ then find value of $\frac{4\epsilon_{r_2}}{\epsilon_{r_1}}$. ($\mu_r$ \& ϵ_r are relative permeability \& permittivity of medium)

Q.26

The following conversion can be carried out by



(A) Red hot Fe, 873K → CH_3COCl, $AlCl_3$ → ZnHg, HCl → Br_2, Fe

(B) Red hot Fe, 873K → Br_2, Fe → CH_3COCl, $AlCl_3$ → ZnHg, HCl

(C) Red hot Fe, 873K → CH_3COCl, $AlCl_3$ → Br_2, Fe → ZnHg, HCl

(D) Red hot Fe, 873K → Br_2, Fe → ZnHg, HCl → CH_3COCl, $AlCl_3$

Q.27

Among the following, which one is a non-reducing sugar?

(A) Glyceraldehyde

(B) Glucose

(C) Fructose

(D) Sucrose

Q.28

Which of the following set of elements are not in accordance with the Law of Octave rule?

(A) H, F, Cl

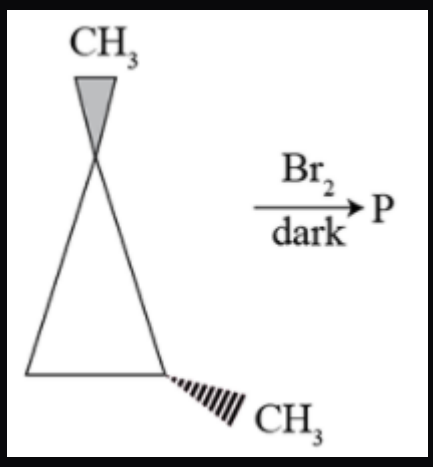
(B) Li, Na, K

(C) C, Si, Ge

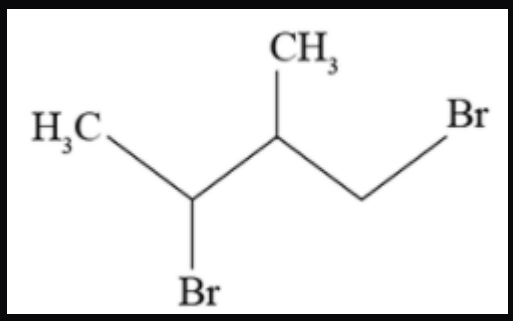
(D) Be, Mg, Ca

Q.29

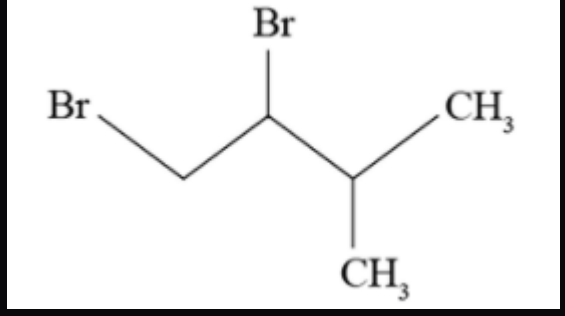
The major product P of the following reaction is



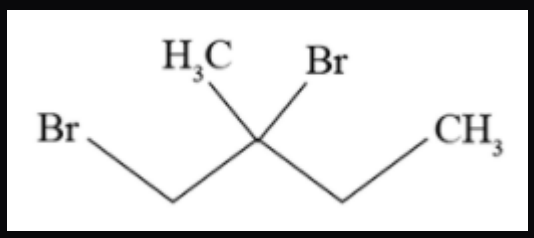
(A)



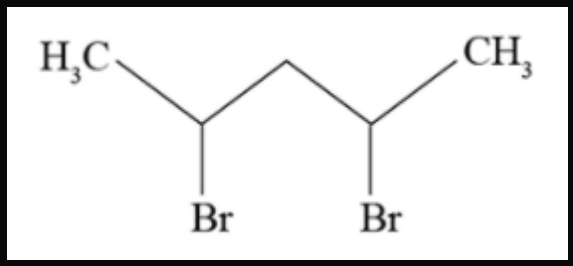
(B)



(C)



(D)



Q.30

Assertion : If x mL of a $NaHCO_3$ solution requires y mL of HCI, to reach the end point, same volume of the stock solution would require the same y mL of NaOH, whose molarity is same as that of HCl.
Reason : $NaHCO_3$ as an amphoteric salt can be neutralized by either a strong acid or a strong base.

(A) Both assertion and reason are correct, but reason is not the correct explanation of assertion.

(B) Assertion is correct, but reason is wrong.

(C) Assertion is wrong, but reason is correct.

(D) Both assertion and reason are correct and reason is the correct explanation of the assertion.

Q.31

Match the list I with list II and select the correct answer using the codes given below the list.

	List I		List II
A.	Marble	1.	Formation of urea
B.	Hydrated CO_2	2.	KOH
C.	CO_2 absorbs	3.	H_2CO_3
D.	Use of CO_2	4.	$CaCO_3$

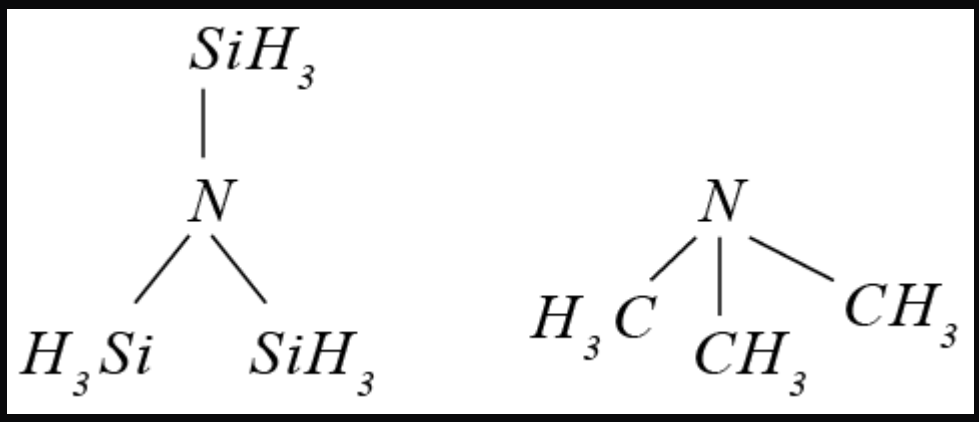
(A)	A	B	C	D
	4	3	2	1
(B)	A	B	C	D
	4	2	3	1
(C)	A	B	C	D
	1	2	3	4

(D)

A	B	C	D
1	3	2	4

Q.32

Trimethylammonia is pyramidal whereas trisilylammonia is a planar molecule. Why?



(A) In silicon atom, the electron pair of nitrogen is involved in $p\pi - d\pi$ bonding.

(B) In nitrogen, lone pair of electrons is present at nitrogen hence it has pyramidal shape.

(C) Both (a) and (b) are correct.

(D) Both (a) and (b) are wrong.

Q.33

If $0.01\mathrm{M}$ solution of an electrolyte has a resistance of $40\mathrm{ohms}$ in a cell having a cell constant of $0.4\ \mathrm{cm}^{-1}$, then its molar conductance in $\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{mol}^{-1}$

(A) 10^2

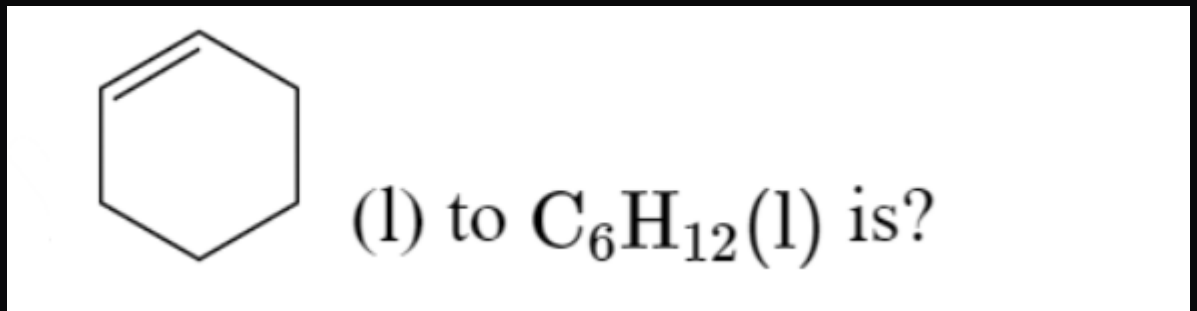
(B) 10^4

(C) 10

(D) 10^3

Q.34

If enthalpy of hydrogenation of $\mathrm{C_6H_6(l)}$ into $\mathrm{C_6H_{12}(l)}$ is $-205\ \mathrm{kJ}$ and resonance energy of $\mathrm{C_6H_6(l)}$ is $-152\ \mathrm{kJ\ mol^{-1}}$ then enthalpy of hydrogenation of



[Assume enthalpiies of vapourisation of all liquids involved are equal].

(A) $-535.5\ \mathrm{kJ/mol}$

(B) $-238\ \mathrm{kJ/mol}$

(C) $-357\ \mathrm{kJ/mol}$

(D) $-119\mathrm{k}J/\mathrm{mol}$

Q.35

Products of the following reaction

$\mathrm{CH_3C \equiv C\ CH_2CH_3} \xrightarrow[\text{(2) Hydrolysis}]{\text{(1) } O_3} \mathrm{A + B}$, A and B are

(A) $\mathrm{CH_3CHO,\ CH_3CH_2CHO}$

(B) $\mathrm{CH_3COOH,\ CH_3CH_2CHO}$

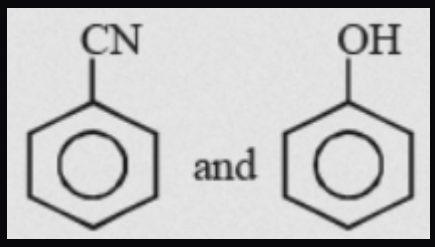
(C) $\mathrm{CH_3COOH,\ HO\,OCCH_2CH_3}$

(D) $\mathrm{CH_3COOH,\ CO_2}$

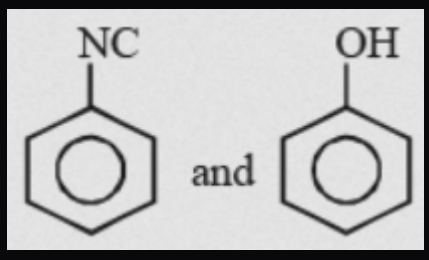
Q.36

A mixture of two aromatic compounds A and B is separated by dissolving in chloroform followed by extraction with aq. KOH solution. The alkaline aqueous layer gives a mixture of two isomeric compounds on treatment with carbon tetrachloride. The organic layer containing compound A gives an unpleasant odour on treatment with alcoholic solution of KOH. Compounds A and B respectively are

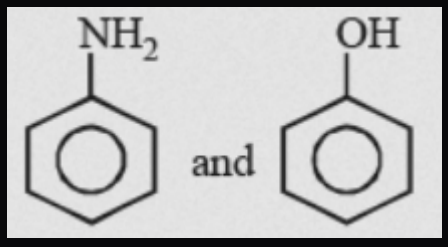
(A)



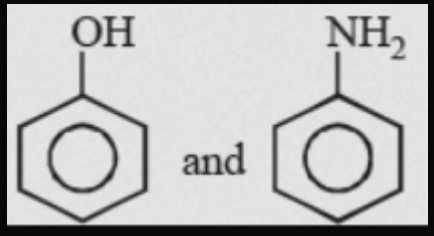
(B)



(C)

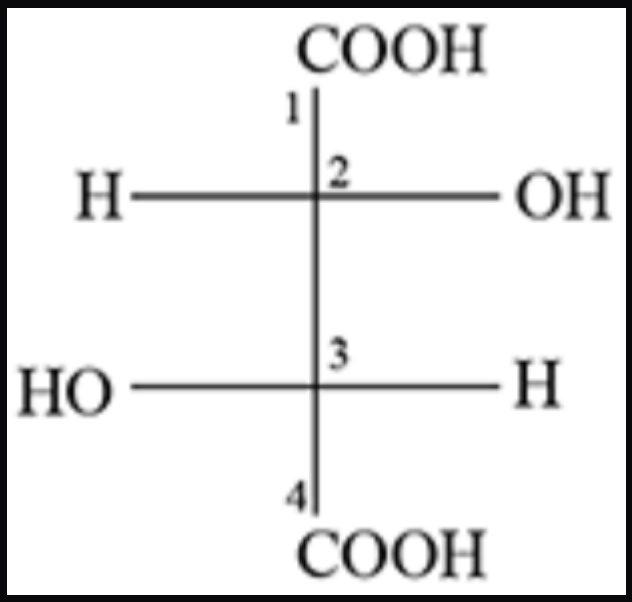


(D)



Q.37

In the compound



Configuration of C_2 and C_3 atoms respectively are

(A) S, S

(B) S, R

(C) R, S

(D) R, R

Q.38

STATEMENT I: A catalyst can increase the rate constant to a large extent.
STATEMENT II: By using suitable catalyst, we can reach reaction completion faster.

(A) Statement I is true, Statement II is true; Statement II is the correct explanation for Statement I.

(B) Statement I is true, Statement II is true; Statement II is not the correct explanation for Statement I.

(C) Statement I is true; Statement II is false.

(D) Statement I is false; Statement II is true.

Q.39

Statement - 1: If a liquid solute which is more volatile than the solvent, is added to it then the vapour pressure may increase.
Statement - 2: In the presence of a more volatile liquid solute, only solute will form the vapours and solvent will not.

(A) Statement-1, is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement-1.

(B) Statement - 1 is True, Statement is True; Statement - 2, is NOT a correct explanation for Statement - 1.

(C) Statement - 1 is True, Statement - 2, is False.

(D) Statement - 1 is False, Statement - 2 is True.

Q.40

What is Z in the following reaction sequence?

$$C_6H_5NH_2 \xrightarrow[\text{(ii)}\ H_3PO_2+H_2O]{\text{(i)}\ NaNO_2+HCl/273\ K} \xrightarrow{\text{(iii)}\ CO,HCl\ \text{anhy}\cdot AlCl_3/CuCl} Z.$$

(A) $C_6H_5CO_2H$

(B) C_6H_5OH

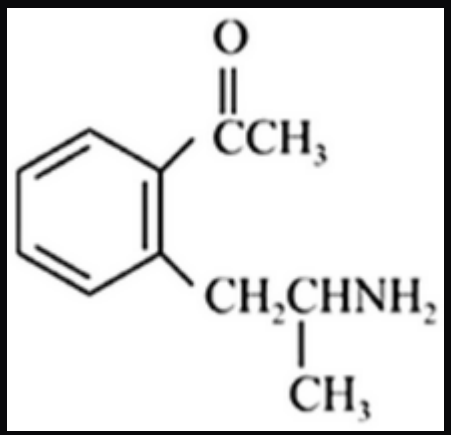
(C) C_6H_5CHO

(D) C_6H_6

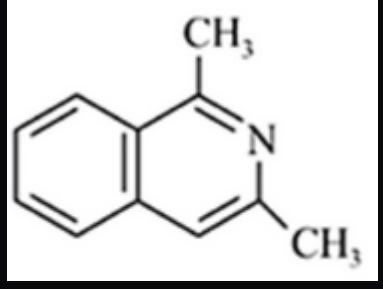
Q.41

The amino ketone shown undergoes a spontaneous cyclisation on standing.

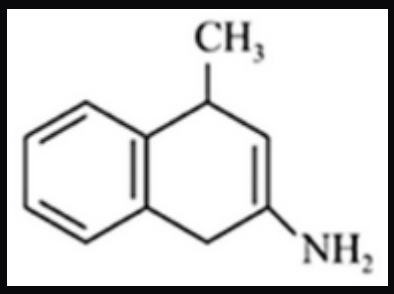
What is the product of this intramolecular reaction?



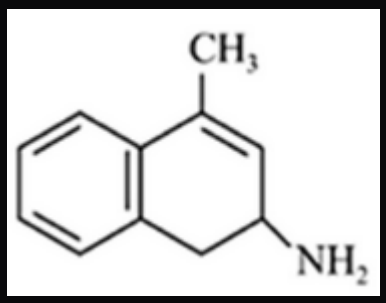
(A)



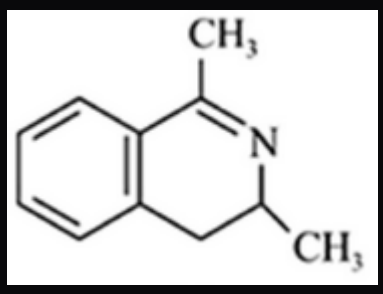
(B)



(C)



(D)



Q.42

If the dipole moment of Toluene and Nitro-benzene are $0.43\mathrm{D}$ and $3.93\mathrm{D}$ respectively, then what is the expected dipole moment of p-Nitrotoluene?

(A) $3.50\mathrm{D}$

(B) $2.18\mathrm{D}$

(C) $4.36\mathrm{D}$

(D) $5.30\mathrm{D}$

Q.43

The hybridization states of the central atoms in the complexes $[\mathrm{Fe(CN)_6}]^{3-}$, $[\mathrm{Fe(CN)_6}]^{4}$ and $[\mathrm{Co\ (NO_2)_6}]^{3-}$ are

(A) d^2sp^3, sp^3 and d^4s^2 respectively

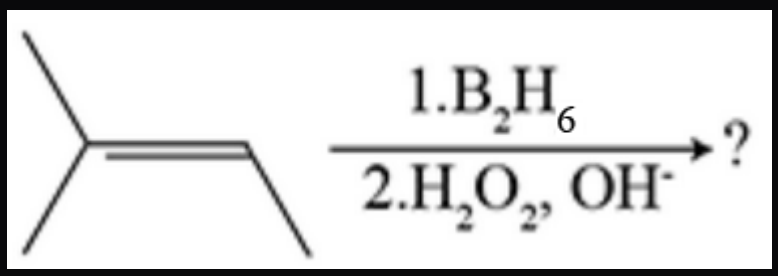
(B) d^2sp^3, sp^3d and sp^3d^2 respectively

(C) d^2sp^3, sp^3d^2 and dsp^2 respectively

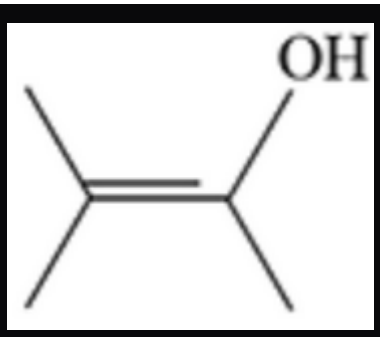
(D) all d^2sp^3

Q.44

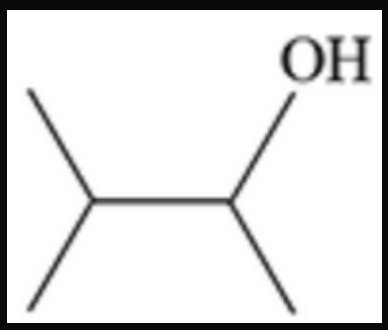
The major product of the following reaction sequence is :



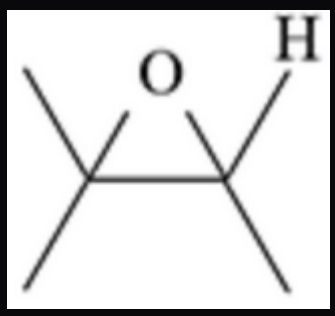
(A)



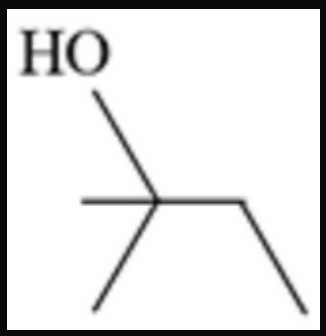
(B)



(C)



(D)



Q.45

RNA and DNA are chiral molecules, their chirality is due to

(A) chiral bases

(B) chiral phosphate ester units

(C) D-sugar component

(D) L-sugar component

Q.46

In the following sequence of reactions the maximum number of atoms present in molecule ' C ' in one plane is

$$\mathrm{A} \xrightarrow[\text{Cu tube}]{\text{Red hot}} \mathrm{B} \xrightarrow[\text{Anhydrous } \mathrm{AlCl_3}]{\mathrm{CH_3Cl}(1eq.)} \mathrm{C}$$

(A is a lowest molecular weight alkyne)

Q.47

Ethylene glycol is used as an antifreeze in cold climate. Mass of ethylene glycol which should be added to $4\ \mathrm{kg}$ of water to prevent it from freezing at $-6^{\circ}\mathrm{C}$ will be ..... g. (K_f for water $= 1.86\ \mathrm{K\ kg\ mol^{-1}}$ and molar mass of ethylene glycol $= 62\ \mathrm{g\ mol^{-1}}$)

Q.48

Find the total number of possible isomers for the complex compound

$$\left[\mathrm{Cu^{II}(NH_3)_4}\right]\left[\mathrm{Pt^{II}Cl_4}\right]$$

Q.49

If each hydrogen atom in the ground state, 1.0 mol of H atoms are excited by absorbing photons of energy 8.4 eV, 12.09 eV, and 15.0 eV of energy, then the number of spectral lines emitted is equal to.

Q.50

A and B decompose via first order kinetics with half-lives $54.0\ \mathrm{min}$ and $18.0\ \mathrm{min}$ respectively. Starting from an equimolar non-reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B in min. (Round off to the Nearest Integer).

Q.51

There are 7 distinguishable rings. The number of possible five-rings arrangements on the four fingers (except the thumb) of one hand (the order of the rings on each finger is to be counted and it is not required that each finger has a ring) is equal to

(A) 214110

(B) 211410

(C) 124110

(D) 141120

Q.52

Let $A,\ B$ and C be three points taken on the parabola $y^2 = 4ax$ with coordinates $\left(at_i^2, 2at_i\right)\ i = 1,\ 2,\ 3$, where $t_1,\ t_2$ and t_3 are in AP. If $AA',\ BB'$ and CC', are focal chords and coordinates of A', B', C' are $\left(at_i'^2, 2at_i'\right),\ i = 1,\ 2,\ 3$ then $t_1',\ t_2'$ and t_3' are in

(A) AP

(B) GP

(C) HP

(D) AGP

Q.53

Let $\phi(\mathrm{x}) = \int_0^1 \mathrm{e}^{\mathrm{x}}\mathrm{e}^{\mathrm{t}}\phi(\mathrm{t})\mathrm{dt} + \mathrm{x}$ and $\phi\left(\ln\left(\mathrm{e}^2 - 3\right)\right)$

is equal to A, then

(A) $A = \ln\left(e^2 - 3\right) - 2$

(B) $\mathrm{A} \in (3, 4)$

(C) $A = e^2 - 3$

(D) $A = \ln\left(e^2 - 3\right) + 2$

Q.54

Let $f(x) = 2\tan^3 x - 6\tan^2 x + 1 + \operatorname{sgn}(e^x),\ \forall x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$. Then the positive difference between the least value and the local maximum value of the function is (where $\operatorname{sgn}(f(x))$ represents the signum function)

(A) 7

(B) 8

(C) 9

(D) 10

Q.55

If $\tan^{-1}\frac{1}{2x+1} + \tan^{-1}\frac{1}{4x+1} = \cot^{-1}\left(\frac{x^2}{2}\right)$, then the number of all possible values of x is/are

(A) 1

(B) 2

(C) 3

(D) 0

Q.56

Let R is a relation defined as $R = \{(1,2),(2,3),(3,4)\}$. The minimum number of ordered pairs which should be added to make relation *'R'* equivalence relation, are

(A) 7

(B) 9

(C) 13

(D) 11

Q.57

The number of distinct real values of λ for which the vectors $\lambda^3\hat{i}+\widehat{k},\ \hat{i}-\lambda^3\hat{j}$ and $\hat{i}+(2\lambda-\sin\lambda)\hat{j}-\lambda\widehat{k}$ are coplanar, is/are

(A) 1

(B) 0

(C) 2

(D) 3

Q.58

Let $f:R\to R$ is a function satisfying

$f(10-x)=f(x)$ and $f(2-x)=f(2+x),\forall x\in R$. If $f(0)=101$. Then, the minimum possible number of values of x satisfying $f(x)=101, x\in[0,25]$ is

(A) 9

(B) 4

(C) 6

(D) 11

Q.59

In a ΔABC, AD bisects angle A and D lies on BC. Length of AD when $\overrightarrow{AB}=2i+2j+k$ and $\overrightarrow{AC}=3i+4k$ is:

(A) $\frac{3}{8}\sqrt{30}$

(B) $\frac{5}{8}\sqrt{30}$

(C) $\frac{1}{8}\sqrt{30}$

(D) $\frac{\sqrt{30}}{4}$

Q.60

The mean and variance of 7 observations are 7 and 22 respectively. If 5 of the observations are $2,4,10,12,14$, then the remaining 2 observations are

(A) $4,3$

(B) $2,5$

(C) $6, 1$

(D) $4, 2$

Q.61

Let $J = \int_0^{e-1} \frac{1}{x+1} \exp \cdot \left(\frac{x^2+2x-1}{2} \right) dx$ and $K = \int_1^e x \ln x \exp \cdot \left(\frac{x^2-2}{2} \right) dx$. The value of $(\mathrm{J} + \mathrm{K})$ is equal to (Here exp.(x) means e power x)

(A) $(\sqrt{\mathrm{e}})^{\mathrm{e}^2+1}$

(B) $(\sqrt{\mathrm{e}})^{\mathrm{e}^2-1}$

(C) 0

(D) $(\sqrt{\mathrm{e}})^{\mathrm{e}^2-2}$

Q.62

Let $x,\ y,\ z$ be positive and $x,\ y,\ z\ \neq 1$. Let $\Delta = \begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$. Then the numerical value of Δ is

(A) -1

(B) 0

(C) 1

(D) None of these

Q.63

Consider the matrix $A = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}$ and B be a square matrix of order 2 such that $BB^T = B^T B = I$. Let $C = BAB^T$ and $D = [d_{ij}]_{2\times 2} = B^T C^6 B$, then d_{11} is equal to

(A) 25

(B) 27

(C) 30

(D) 81

Q.64

If $A\left(\overrightarrow{a}\right), B\left(\overrightarrow{b}\right), C\left(\overrightarrow{c}\right)$ and $D\left(\overrightarrow{d}\right)$ are four points such that $\overrightarrow{a} = -2\hat{i} + 4\hat{j} + 3\hat{k}, \overrightarrow{b} = 2\hat{i} - 8\hat{j}, \overrightarrow{c} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\overrightarrow{d} = 4\hat{i} + \hat{j} - 7\hat{k}$ and d_1 is the shortest distance between the lines AB and CD, then which of the following is true:

(A) $d_1 = 0$, hence AB and CD intersect

(B) AB and CD are skew lines and $d_1 = \frac{20}{13}$

(C) AB and CD are skew-lines and $d_1 = \frac{23}{13}$

(D) *AB* and *CD* are skew lines and $d_1 = \frac{25}{13}$

Q.65

If $\mathrm{z}_1, \mathrm{z}_2, \mathrm{z}_3$ are three points lying on the circle $|\mathrm{z}| = 2$ then the minimum value of $|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2$ is equal to

(A) 6

(B) 12

(C) 15

(D) 24

Q.66

If some three consecutive coefficients in the binomial expansion of $(x+1)^n$ in powers of x are in the ratio $2 : 15 : 70$, then the average of these three coefficients is:

(A) 227

(B) 964

(C) 625

(D) 232

Q.67

If the area bounded by $f(x) = \tan^3 x + \tan x$ from $x = 0$ to $x = \frac{\pi}{4}$ is k square units, then the maximum value of $g(x) = k \sin x$ is $\left(\forall x \in \left[0, \frac{\pi}{4}\right]\right)$

(A) 2

(B) $\frac{1}{2\sqrt{2}}$

(C) 4

(D) 8

Q.68

The solution of the equation $\frac{dy}{dx} + y \tan x = x^m \cos x$ is

(A) $(m+1)y = x^{m+1} \cos x + c(m+1) \cos x$

(B) $my = (x^m + c) \cos x$

(C) $y = \left(x^{m+1} + c\right) \cos x$

(D) None of these

Q.69

Two numbers 'a' and 'b' are chosen at random from the numbers 1, 2, 3,....30. The chance that $a^2 - b^2$ is divisible by '3' is

(A) $\frac{9}{87}$

(B) $\frac{12}{87}$

(C) $\frac{15}{87}$

(D) $\frac{47}{87}$

Q.70

Let triangle ABC is right-angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If $BD = 20$ units and $DC = 16$ units, then the length AB is equal to

(A) $6\sqrt{21}$ units

(B) $6\sqrt{10}$ units

(C) $\sqrt{300}$ units

(D) 18 units

Q.71

The sum of all the values of $|\lambda|$ such that the lines $x + 2y - 3 = 0$, $3x - y - 1 = 0$ and $2\lambda x + y - 2 = 0$ cannot form a triangle is K, find 100K

Q.72

The remainder when $\left(\sum_{k=1}^{5} {}^{20}\mathrm{C}_{2k-1}\right)^6$ is divided by 11 , is :

Q.73

If $6p^2$ numbers of 4 digits can be made with the digit 0, 1, 2, 3, 4, 5, which are divisible by 3, digit being unrepeated in the same number, then find the value of p + 4.

Q.74

If z is a complex number such that $|z| = 2$, then the area (in sq. units) of the triangle whose vertices are given by $z, -iz$ and $iz - z$ is equal to

Q.75

Let three positive numbers $a,\ b,\ c$ are in geometric progression, such that $a,\ b + 8,\ c$ are in arithmetic progression and $a,\ b + 8,\ c + 64$ are in geometric progression. If the arithmetic mean of a, b, c is k, then $\frac{3}{13}k$ is equal to

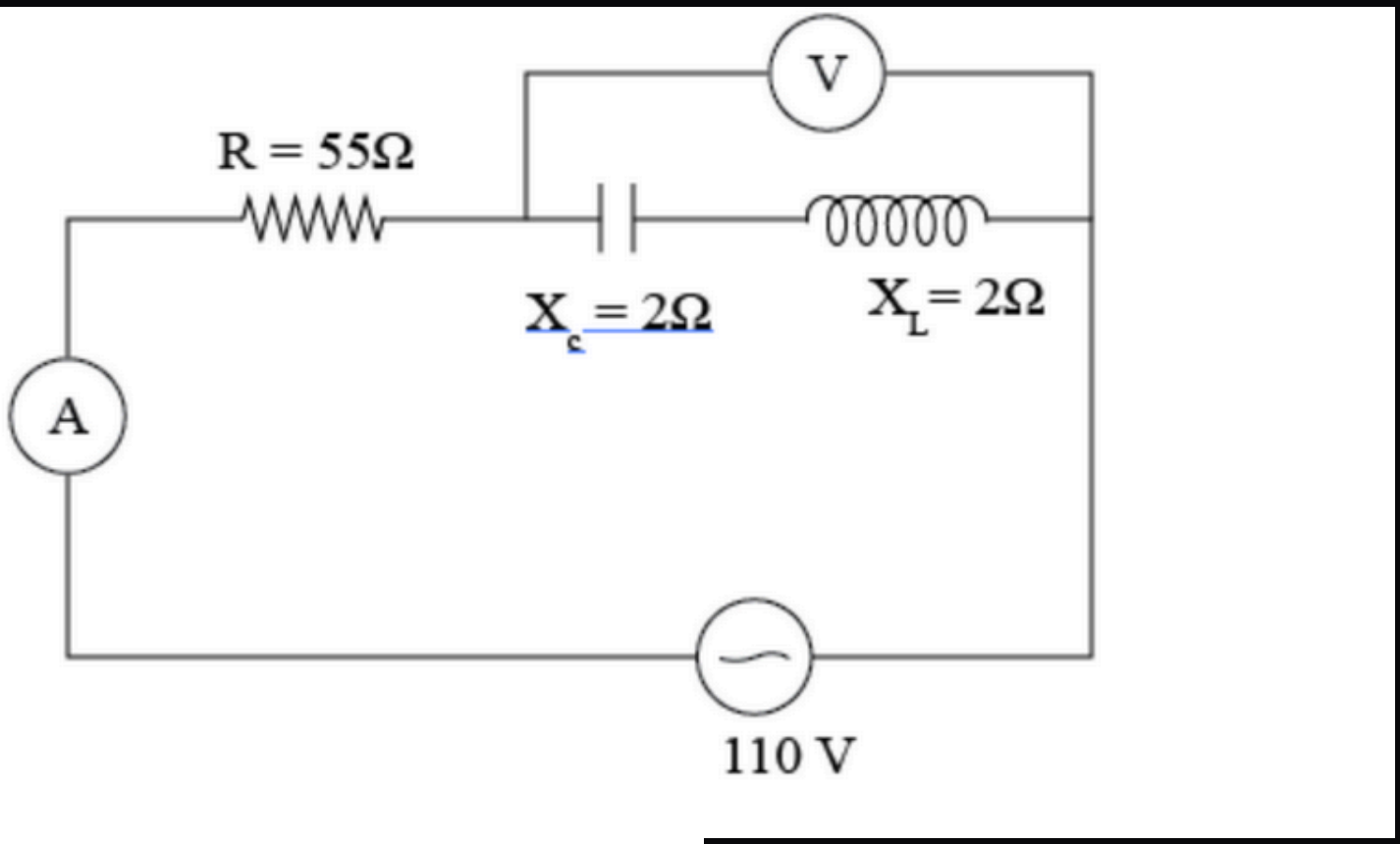
Answers & Solutions

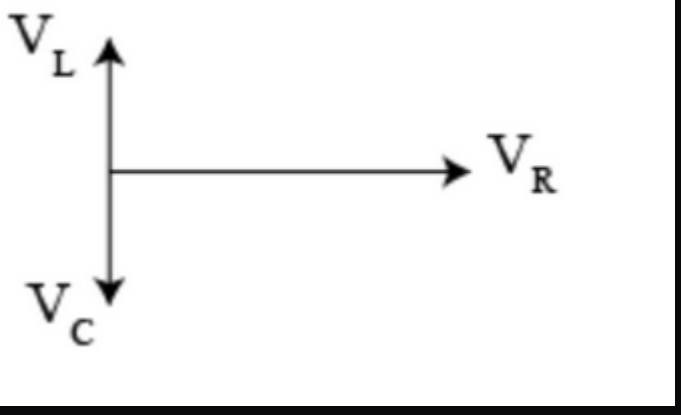
Q.1 Answer:

2 A, 0 V

Solution:

The circuit is RLC resonant circuit.





$\therefore$ **Reading of voltmeter** $= V_L - V_C = 0$

Reading of ammeter $= \frac{E_{\text{rms}}}{Z} = \frac{E_{\text{rms}}}{R} = \frac{110}{55} = 2\ \text{A}$

Q.2 Answer:

$g\left(\frac{M-m}{M+m}\right)^2$ downward

Solution:

$a_{com} = \frac{m_1\bar{a}_1+m_2\overline{a_2}}{m_1+m_2}$

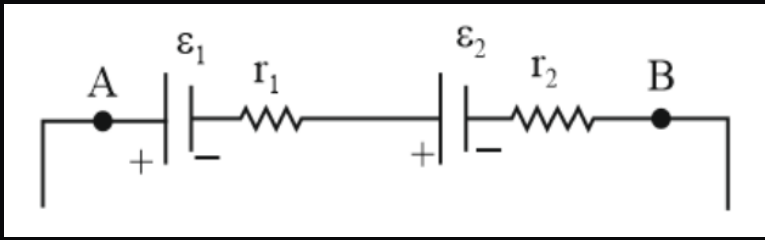
$a_{com} = \frac{1}{m+M}\left[\frac{M(M-m)}{m+M} - \frac{m(M-m)}{m+M}\right]g$

$= \frac{1}{m+M}\left[\frac{(M-m)^2}{m+M}\right]g = \left(\frac{M-m}{M+m}\right)^2 g$

Q.3 Answer:

Each of A and B is wrong.

Solution:



Two batteries having resistance ε_1 & ε_2 are connected in series having an internal resistance r_1 & r_2, then equivalent emf should be,

$\varepsilon_{eff} = \varepsilon_1 + \varepsilon_2$ but if one battery is reversed then $E_{eff} = E_1 - E_2$ and equivalent resistance also, $r_{eff} = r_1 + r_2$, it means both parameter having larger value either of two.

Q.4 Answer:

$\left[\text{V}^{-4}\text{A}^2\text{F}\right]$

Solution:

Let $Y = f(V,\ F,\ A)$

$Y = KV^xF^yA^z,\ K \rightarrow$ **Unit less**

$[Y] = [V]^x[F]^y[A]^z$

$\left[\text{ML}^{-1}\,\text{T}^{-2}\right] = \left[\text{LT}^{-1}\right]^{\text{x}}\left[\text{MLT}^{-2}\right]^{\text{y}}\left[\text{LT}^{-2}\right]^{\text{z}}$

$[\text{M}]\left[\text{L}^{-1}\right]\left[\text{T}^{-2}\right] = [\text{M}]^{\text{y}}\left[\text{L}^{\text{x+y+z}}\right]\left[\text{T}^{-\text{x}-2\text{y}-2\text{z}}\right]$

$\Rightarrow \quad y = 1,\ x + y + z = -1;\ \ -x - 2y - 2z = -2$

$x + z = -2 \ \ x + 2y + 2z = 2$

$z = 2,\ x = -4 \ \ x + 2z = 0$

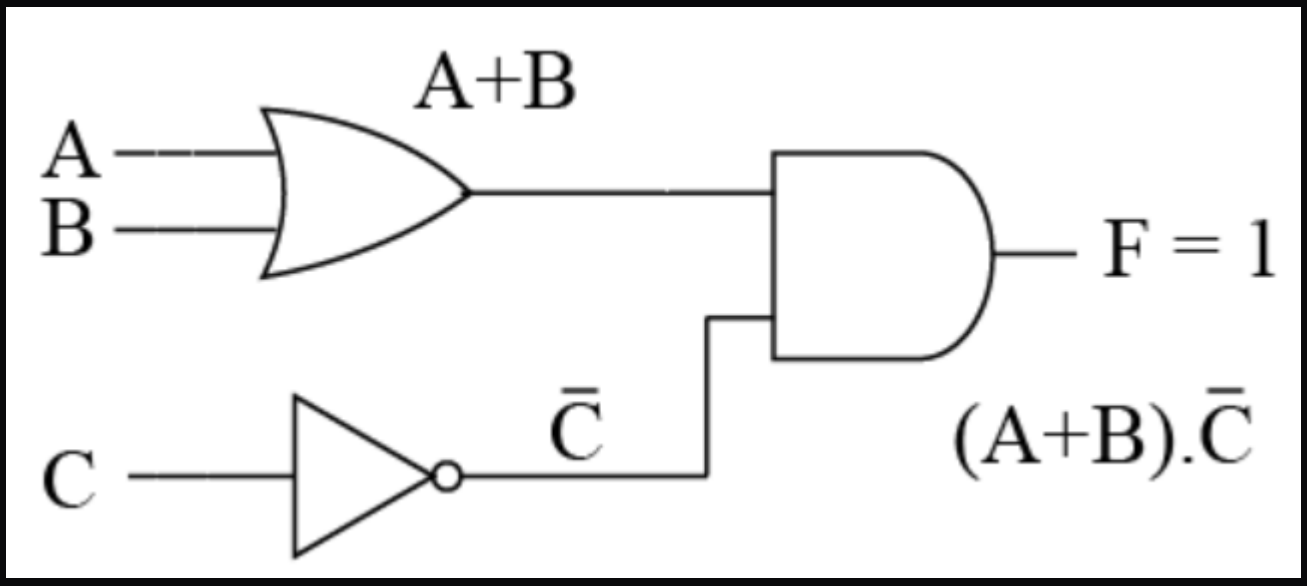
$|\mathrm{Y}| = \left[\mathrm{V}^{-4}\mathrm{A}^{2}\mathrm{F}\right]$

Q.5 Answer:

$A = 0, B = 1, C = 0$

Solution:

When $A = 0, B = 1$ and $C = 0$, then output becomes 1.



Q.6 Answer:

zero

Solution:

Pull on the rope = effective weight

$= [1650 + (1500 \times 0.2) - 1500 \times 1.3]\ \mathrm{kgf}$

$= 1650 + 300 - 1950 = 0$

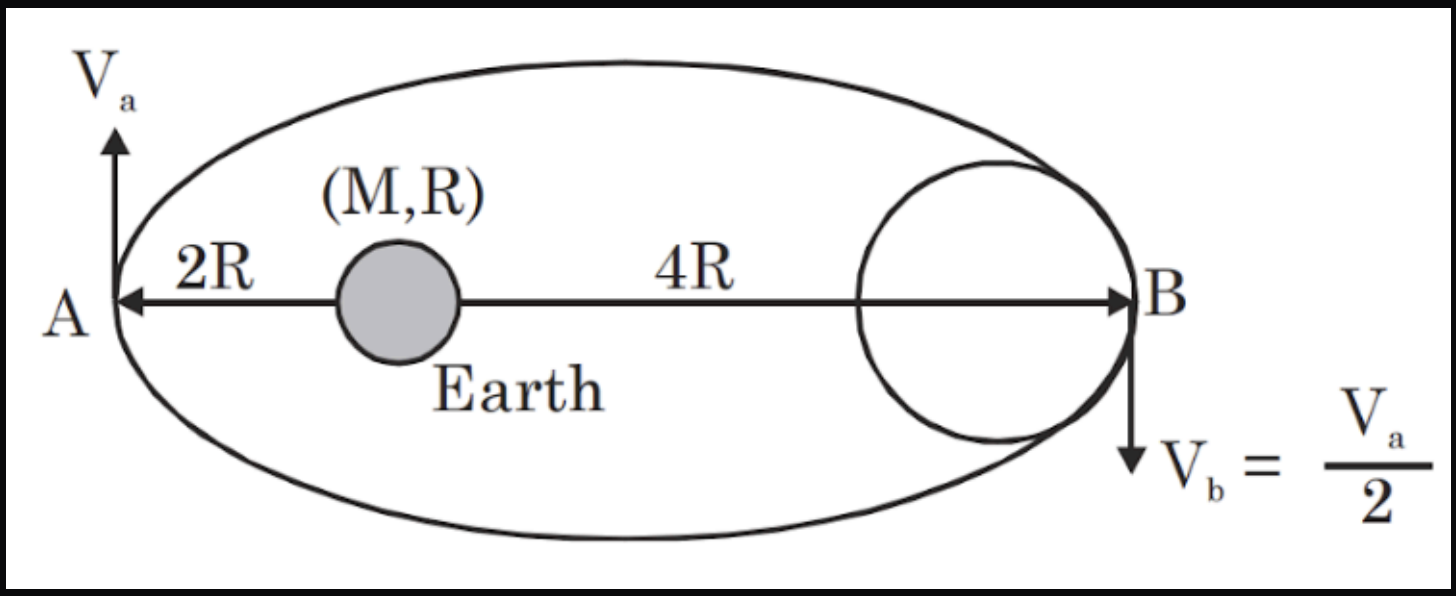
Q.7 Answer:

$\frac{8R}{3}$

Solution:

$$\frac{1}{2}mV_a^2 - \frac{GMm}{2x} = \frac{1}{2}m\frac{V_a^2}{4} - \frac{GMm}{4x},$$

$$V_a = \sqrt{\frac{2}{3}\frac{GM}{x}}$$



ROC at point - B

$$ROC = \frac{V_b^2}{a_\perp} = \frac{\frac{2GM}{3R} \times \frac{1}{4}}{\frac{GM}{(4R)^2}} = \frac{8R}{3}$$

Q.8 Answer:

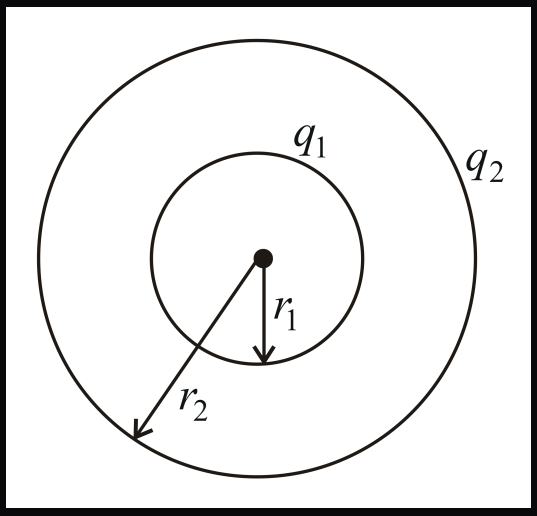
Assertion is true, reason is true. Reason is a correct explanation for assertion.

Solution:

Assertion: Two concentric charged spherical shells are given.

Let q_1 and q_2 are the charges on concentric spherical shells of radii r_1 and r_2 respectively as shown in figure.

Here, $r_2 > r_1$



The electric potential of inner spherical shell is,

$$V_1 = \frac{q_1}{4\pi\varepsilon_0 r_1} + \frac{q_2}{4\pi\varepsilon_0 r_2}$$

The electric potential of outer spherical shell is,

$$V_2 = \frac{q_1}{4\pi\varepsilon_0 r_2} + \frac{q_2}{4\pi\varepsilon_0 r_2}$$

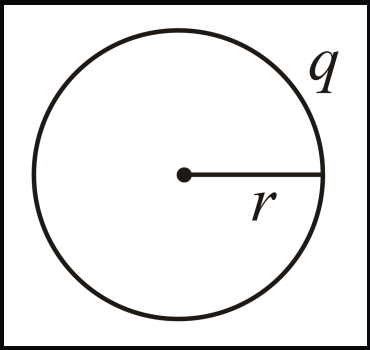
$\therefore$ The potential difference between the spherical shell is,

$$V_1 - V_2 = \frac{q_1}{4\pi\varepsilon_0 r_1} + \frac{q_2}{4\pi\varepsilon_0 r_2} - \frac{q_1}{4\pi\varepsilon_0 r_2} - \frac{q_2}{4\pi\varepsilon_0 r_2}$$

$$\Rightarrow V_1 - V_2 = \frac{q_1}{4\pi\varepsilon_0}\left[\frac{1}{r_1} - \frac{1}{r_2}\right]$$

Here potential difference depends on the inner shell i.e., (q_1)

Reason: Let q be the charge on the spherical shell of radius r.



The electric potential of the spherical surface $V = \frac{q}{4\pi\varepsilon_0 r}$

$\because$ **Electric field** $E = -\frac{\mathrm{d}V}{\mathrm{d}r}$

Inside the spherical shell electric field $E = 0$

$\Rightarrow 0 = -\frac{dV}{dr}$

$V =$**constant**

It means the electric potential at every point inside the shell is the same as that on the surface.

Q.9 Answer:

zero

Solution:

As the material of the loop is same, resistance through the path with 3 sides will be 3 times the resistance through the path with one side. So current will get divided in the ratio shown in the figure.
Also, θ_1 **and** θ_2 **mentioned in the formula will be equal to** 45° **as the loop is square loop. (can be deduced from the figure)**
The magnetic field due to the shorter path will be $\frac{\mu_0 \cdot 0.75\mathrm{I}}{4\pi \cdot \frac{a}{\sqrt{2}}}$
The magnetic field due to the longer path will be $3 \times \frac{\mu_0 \cdot 0 \cdot 25\mathrm{I}}{4\pi \cdot \frac{\mathrm{a}}{\sqrt{2}}} = \frac{\mu_0 \cdot 0.75\mathrm{I}}{4\pi \cdot \frac{\mathrm{a}}{\sqrt{2}}}$
By Ampere's right hand grip rule, we can conclude that the magnetic field due to both these paths is opposite in direction. So, the fields will cancel out and the resultant field at the center will be zero.

Q.10 Answer:

8.55 MeV

Solution:

The nucleus $^{40}_{20}\text{Ca}$ **contains 20 protons and 20 neutrons.**

Mass of 20 protons $= 20 \times 1.007825$

$= 20.1565\,u$

Mass of 20 neutrons $= 20 \times 1.008665$

$= 20.1733\,u$

Total mass $= 40.3298\,u$

Mass $^{40}_{20}\text{Ca}$ **nucleus** $= 39.962589\,u$

Mass defect, Δm $= 0.367211\,u$

$\text{Binding energy} = 0.367211 \times 931 = 341.87 \text{ MeV}$

$\text{Binding energy per nucleon} = \frac{341.87}{40} = 8.547 \text{ MeV}$

Q.11 Answer:

$\alpha = \sin^{-1}\left(\frac{4}{5}\right) \approx 53^\circ$

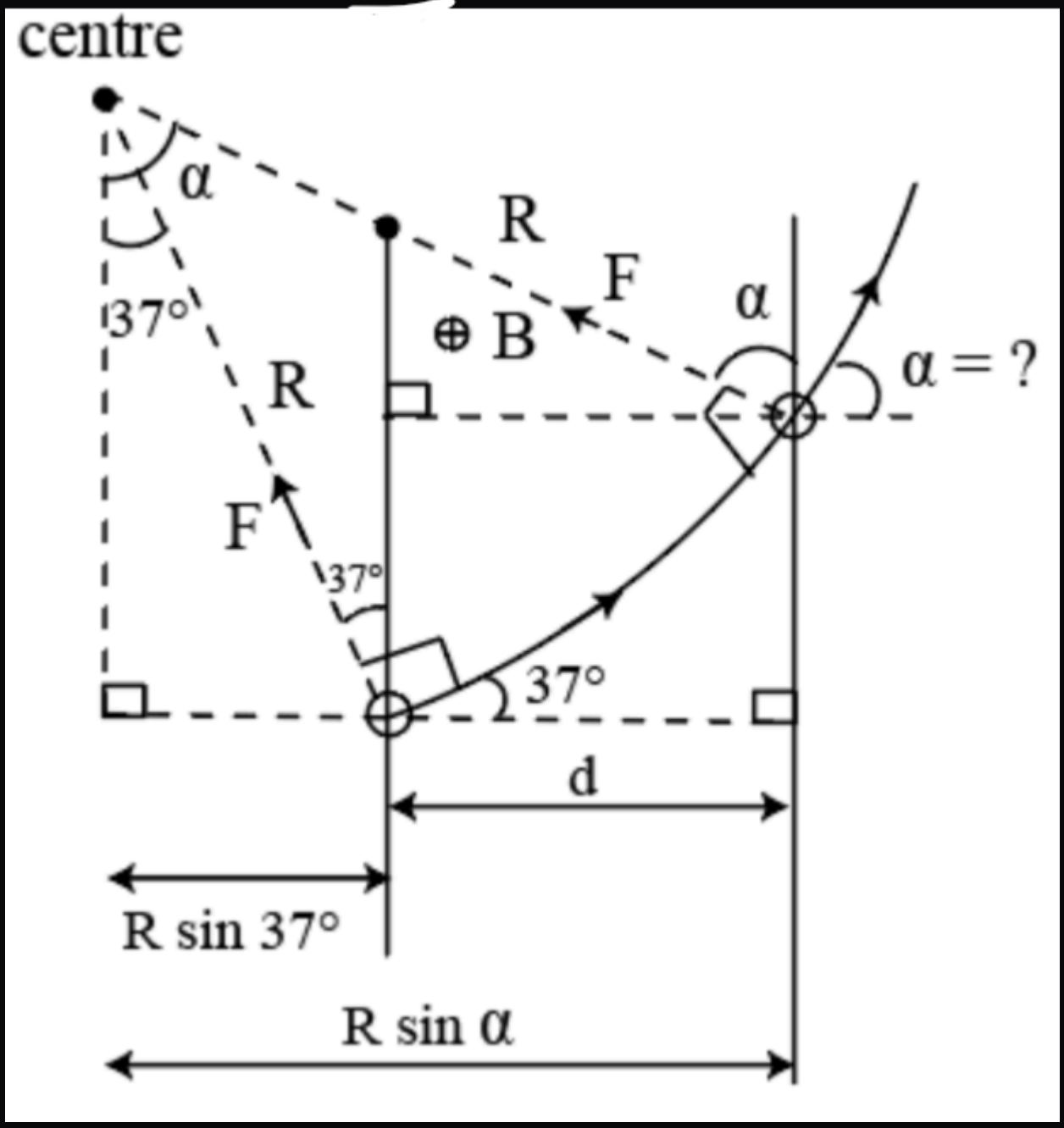
Solution:

$R = \frac{mv}{qB} = \frac{\sqrt{2mqV}}{qB} = \frac{\sqrt{2mV}}{\sqrt{q}B} \quad \because d = R\sin\alpha - R\sin 37^\circ$

$\Rightarrow \frac{\sqrt{2mV}}{5\sqrt{q}B} = \frac{\sqrt{2mV}}{\sqrt{q}B}(\sin\alpha - \sin 37^\circ)$

$\Rightarrow \frac{1}{5} = \sin\alpha - \frac{3}{5} \Rightarrow \sin\alpha = \frac{4}{5}$

$\therefore \alpha = \sin^{-1}\left(\frac{4}{5}\right) \approx 53^\circ$



Q.12 Answer:

i. → a.,b.,c. ii. → a.,b. iii. → a.,b. iv. → a.,b.,c.

Solution:

Electrostatic force and gravitational force are conservative, action-reaction forces. Also electrostatic force depends upon the nature of the medium between interacting objects. The principle of superposition is applicable, irrespective of the nature of force.

Q.13 Answer:

$35.7^{\circ}\mathrm{C}$

Solution:

$\frac{t+5}{105+5} = \frac{37-0}{100-0} \Rightarrow t = 35.7^{\circ}\mathrm{C}$

Q.14 Answer:

Statement - 1 is True, Statement - 2 is False.

Solution:

Number of beats $= 524 - 410 = 114$ Due to persistence of hearing we can hear more than 10 beats per second. two waves of same frequencies and amplitude superimpose give stationary waves.

Q.15 Answer:

1:4

Solution:

Root mean square velocity

$v_{rms} \propto \frac{1}{\sqrt{M}}$

So $\frac{(v_{rms})_{O_2}}{(v_{rms})_{H_2}} = \sqrt{\frac{M_{H_2}}{M_{O_2}}}$

$= \sqrt{\frac{2}{32}} = \frac{1}{4}$

Q.16 Answer:

$\frac{f-a}{2(1+b)}$

Solution:

$V_p = \frac{dx_p}{dt} = a + 2bt$

and $V_Q = \frac{dx_Q}{dt} = f - 2t$**, Given,** $V_p = V_Q$

$\therefore a + 2bt = f - 2t \Rightarrow t = \frac{f-a}{2(b+1)}$

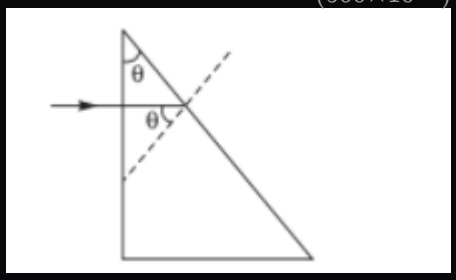
Q.17 Answer:

Light of $\lambda = 400\ \text{nm}$ undergoes total internal reflection

Solution:

We have, $\mu_1 = 1.20 + \frac{0.8 \times 10^{-14}}{(400 \times 10^{-9})^2}$
or $\mu_1 = 1.20 + \frac{0.8 \times 10^{-14}}{400 \times 400 \times 10^{-18}}$
or $\mu_1 = 1.20 + \frac{0.8}{16}$
or $\mu_1 = 1.20 + 0.05$
or $\mu_1 = 1.25$
or $\sin i_c = \frac{1}{1.25} = 0.8$
$\Rightarrow \quad i_c = 53.13^\circ$

Again, $\mu_2 = 1.20 + \frac{0.8\times10^{-14}}{(500\times10^{-9})^2}$



or $\mu_2 = 1.20 + \frac{0.8}{25}$
or $\mu_2 = 1.20 + 0.32$
or $\mu_2 = 1.232$
or $\sin i_c = \frac{1}{1.232} = 0.81$
$\Rightarrow \quad i_c = \sin^{-1} 0.81 = 54.26^\circ$
Now, $\sin\theta = 0.8$ or $\theta = 53.13^\circ$
This angle is clearly greater than critical angle corresponding to wavelength $400\ \text{nm}$. So, light of $400\ \text{nm}$ wavelength undergoes total internal reflection.

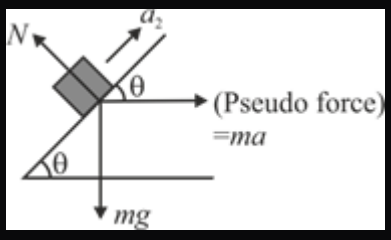
Q.18 Answer:

$\gamma_1 = \gamma_2$ and $I_1 \neq I_2$
Solution:

Here in the graph we can see that, the Stopping potential is same for 1 and 2 . So, frequencies will be same i.e. $y_1 = y_2$ and, currents are different. So, intensity are different i.e. $I_1 \neq I_2$.

Q.19 Answer:

$\frac{1}{\sqrt{5}}$ s
Solution:



Free body diagram of $m\ m\ w.r.t.$ wedge, relative acceleration along the inclined plane

$$a_{\mathrm{r}} = \frac{ma\cos\theta - mg\sin\theta}{m}$$
$$= a\cos\theta - g\sin\theta$$
$$= (10\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) - (10)\left(\frac{1}{2}\right)$$
$$= 10\ \mathrm{m\ s^{-2}}$$
$$t = \sqrt{\frac{2s}{a_{\mathrm{r}}}} = \sqrt{\frac{2\times 1}{10}} = \frac{1}{\sqrt{5}}\ \mathrm{s}$$

$= a\cos\theta - g\sin\theta$

$= (10\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) - (10)\left(\frac{1}{2}\right)$

$= 10\ \mathrm{m\ s^{-2}}$

$t = \sqrt{\frac{2s}{a_{\mathrm{r}}}} = \sqrt{\frac{2\times 1}{10}} = \frac{1}{\sqrt{5}}\ s$

Q.20 Answer:

12

Solution:

In diffraction, angular width of central maxima is $\frac{2\lambda}{b}$ where b is slit width.

In Young's double slit experiment, angular width of each fringe is $\frac{\lambda}{d}$ where d is the distance between the two slits.

Given: $d = 6.1b$

So, number of maximas observed in central maxima of diffraction pattern $\mathrm{n} =$

$\frac{\frac{2\lambda}{b}}{\frac{\lambda}{d}}$

$\therefore \mathrm{n}=\frac{2 \mathrm{d}}{\mathrm{b}}=2 \times 6.1=12.2$

Thus 12 maximas are observed.

Q.21 Answer:

—

Solution:

Let the perpendicular from 0 to the line charge intersects the line charge at point $Q \cdot \mathrm{OQ}=\frac{\sqrt{3} \mathrm{a}}{2}$ **Angle subtended by** AB **at** O **is** $\angle \mathrm{AQB}=2 \angle \mathrm{AQO} \tan \angle \mathrm{AQO}=\frac{\mathrm{AO}}{QO}=\frac{\frac{a}{2}}{\frac{\sqrt{3} a}{2}}=\frac{1}{\sqrt{3}}=\tan 30^{\circ}$

$\therefore \angle \mathrm{AQB}=60^{\circ}$

Let us consider a cylindrical gaussian surface with the axis along line charge of length L **and radius** a **(since** $AQ=a$**)**

Flux through the rectangle is same as the flux through $\frac{1}{6}$ **th of the cuboid since it subtends** 60° **at** O **and field lines are symmetrically radially outwards.**

Hence, $\phi_{\text{rectangle }}=\frac{1}{6} \frac{\lambda}{\epsilon_0}$

$\mathrm{n}=6.$

Q.22 Answer:

—

Solution:

Given : No. of turns across primary coil, N_{p} **= 50**
Number of turns across secondary coil, N_{s} **= 1500**
Magnetic flux linked with primary coil, $\phi=\phi_0+4 \mathrm{t}$

$\therefore$ **Voltage across the primary coil,**

$V_{\mathrm{p}}=\frac{\mathrm{d} \phi}{\mathrm{d} t}=\frac{\mathrm{d}}{\mathrm{d} t}\left(\phi_0+4 t\right)=4 \text{ volt}$

Also, $\frac{V_{\mathrm{s}}}{V_{\mathrm{p}}}=\frac{N_{\mathrm{s}}}{N_{\mathrm{p}}}$

$\therefore V_{\mathrm{s}}=\left(\frac{1500}{50}\right) \times 4=120 \mathrm{~V}$
$\Rightarrow \mathrm{k}=2$

Q.23 Answer:

—

Solution:

$$A_{\text{resultant}} = \sqrt{A_1^2 + A_2^2 + 2\,A_1\,A_2\cos\phi}$$

$$\Rightarrow \sqrt{3}\,A = \sqrt{A^2 + A^2 + 2\,A^2\cos\phi}$$

$$\Rightarrow 3\,A^2 = 2\,A^2 + 2\,A^2\cos\phi$$

$$\Rightarrow \cos\phi = \frac{1}{2}$$

$$\therefore \phi = 60^\circ$$

$$\therefore \text{ Phase difference } = 60 \text{ degree}$$

Q.24 Answer:

—

Solution:

$$\textbf{Least count} = \frac{\text{Pitch}}{\text{Number of division on circuit scale}} = \frac{0.5}{50}$$

$$= 0.01 \text{ mm}$$

$$\textbf{Now, diameter of ball} = (2 \times 0.5) + (2.5 - 5)(0.01) = 1.2 \text{ mm}$$

Q.25 Answer:

—

Solution:

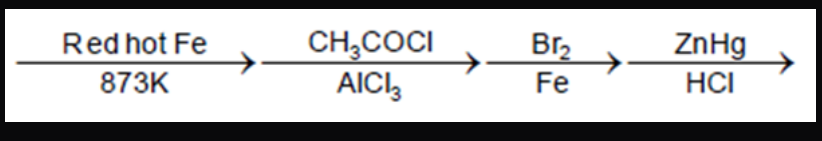
$$V_1 = \frac{\omega}{k} \quad V_2 = \frac{\omega}{3k}$$

$$\Rightarrow 3\,V_2 = V_1$$

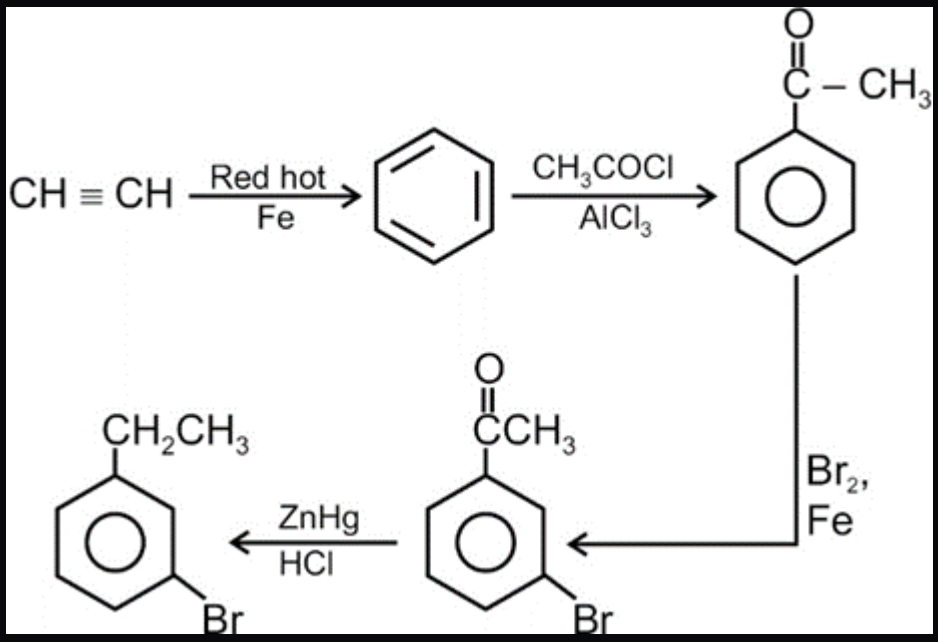
$$3\frac{1}{\sqrt{\mu_{r_2}\epsilon_{r_2}}} = \frac{1}{\sqrt{\mu_{r_1}\epsilon_{r_1}}}$$

$$\Rightarrow \frac{\epsilon_{r_2}}{\epsilon_{r_1}} = 9\frac{\mu_{r_1}}{\mu_{r_2}} = \frac{9}{4} = 2.25$$

Q.26 Answer:



Solution:



Q.27 Answer:

Sucrose

Solution:

Sucrose is not a reducing sugar because it will not reduce Fehling's solution; it does not form an oxime or an osazone, and does not undergo mutarotation. This indicates that neither the aldehyde group of glucose nor the ketonic group of fructose is free in sucrose.

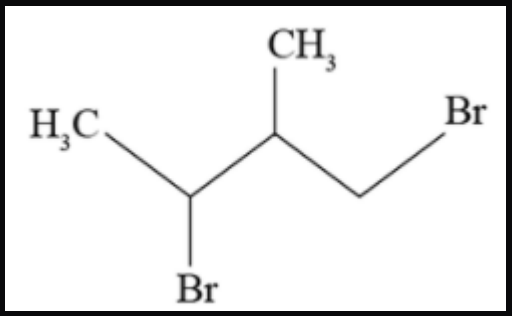
Q.28 Answer:

C, Si, Ge

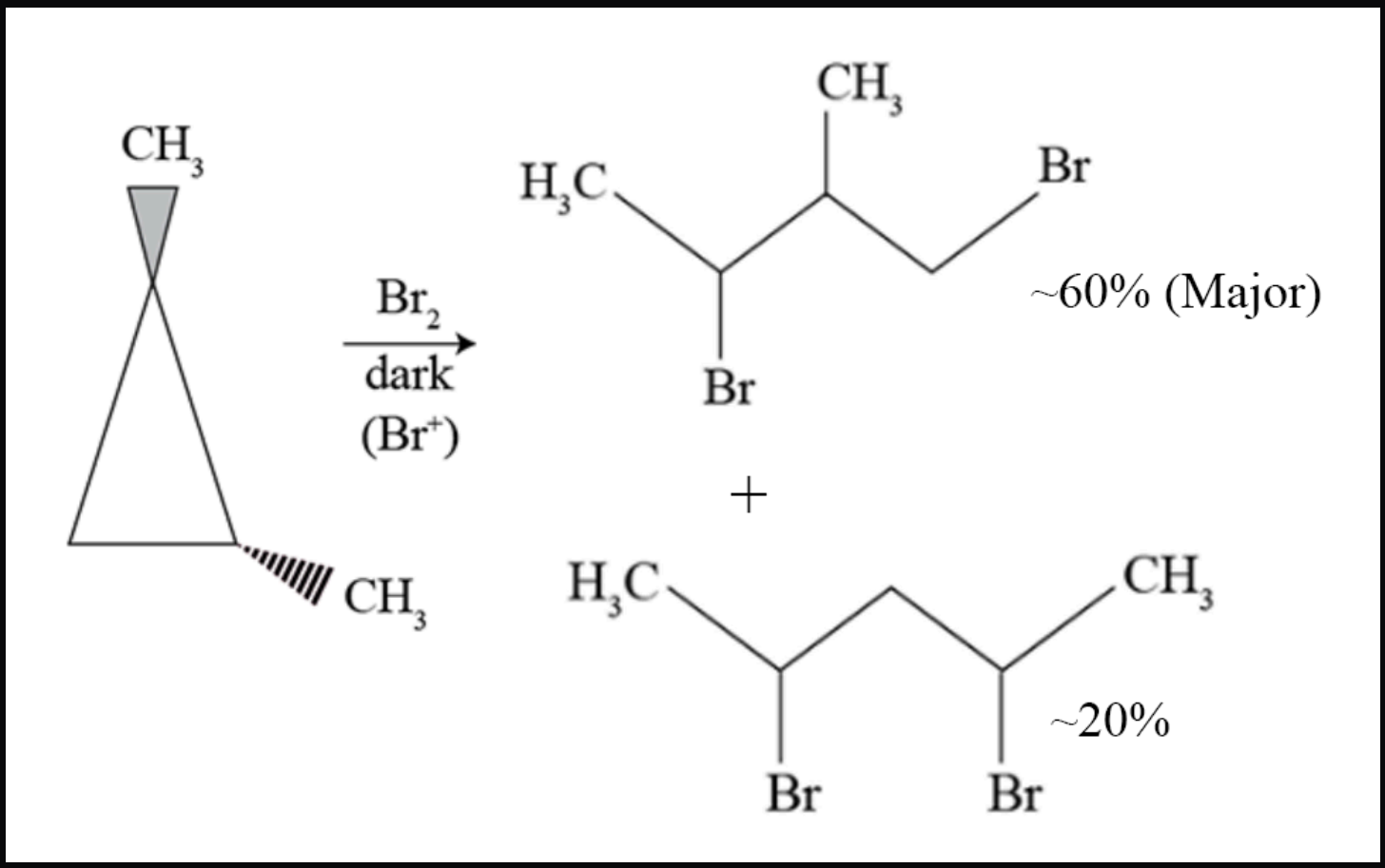
Solution:

Only H to Ca are arranged in Newland's octave and Ge is beyond of 'Ca'

Q.29 Answer:



Solution:



Q.30 Answer:

Both assertion and reason are correct and reason is the correct explanation of the assertion.

Solution:

x mL $NaHCO_3$ <≡> y mL of HCl (N) for end point

x mL $NaHCO_3$ <≡> y mL of NaOH (N) for end point

$NaHCO_3 + HCl \longrightarrow NaCl + H_2CO_3$

$NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$

because $NaHCO_3$ an amphoteric salt it can react with strong acid or strong base and valency factor of HCl, NaOH and $NaHCO_3$ are same = 1.

Both statements are correct.

Q.31 Answer:

A	B	C	D
4	3	2	1

Solution:

Marble is made up of $CaCO_3$.

Carbon dioxide on hydration gives carbonic acid. $\text{Hydrated } CO_2 - H_2CO_3 - (OH_2)CO_2$

CO_2 Is absorbed by KOH Which results in the formation of K_2CO_3.

Carbon dioxide on reacting with ammonia at high temperature gives urea.

$CO_2 + 2\ NH_3 \rightarrow NH_2CONH_2 + H_2O$

Q.32 Answer:

Both (a) and (b) are correct.

Solution:

In silicon atom, there is a vacant d-orbital hence electron pair of nitrogen is donated to silicon by $p\pi - d\pi$ bonding in trisilylammonia and gets sp^2 hybridized hence, it has a planar structure. In trimethyl ammonia, the shape is pyramidal due to presence of lone pair of electrons at nitrogen.

Q.33 Answer:

10^3

Solution:

Molarity $= 0.01\mathrm{M}$; **Resistance** $= 40\mathrm{ohm}$

Cell constant $\left(\frac{l}{A}\right) = 0.4\ \mathrm{cm}^{-1}$

Specific conductivity $(\kappa) = \frac{\text{cell constant}}{\text{resistance}}$

$= \frac{0.4}{40} = 0.01\mathrm{ohm}^{-1}\ \mathrm{cm}^{-1}$

Molar conductance $(\Lambda_{\mathrm{m}}) = \frac{1000\kappa}{\text{Molarity}}$

$= \frac{1000 \times 0.01}{.01} = 10^3\mathrm{ohm}^{-1}\ \mathrm{cm}^2\ \mathrm{mol}^{-1}$

Q.34 Answer:

$-119\mathrm{k}J/\mathrm{mol}$

Solution:

$\mathrm{C_6H_6}(l) + 3\mathrm{H_2}(\mathrm{g}) \longrightarrow \mathrm{C_6H_{12}}(l)$

$\Delta\mathrm{H}^{\circ}_{\mathrm{hyd}(\mathrm{C_6H_6})l} = -205\ \mathrm{kJ}$

Benzene on reactant side so add resonance energy of benzene with sign.

Heat evolved for hydrogenation of three double bonds

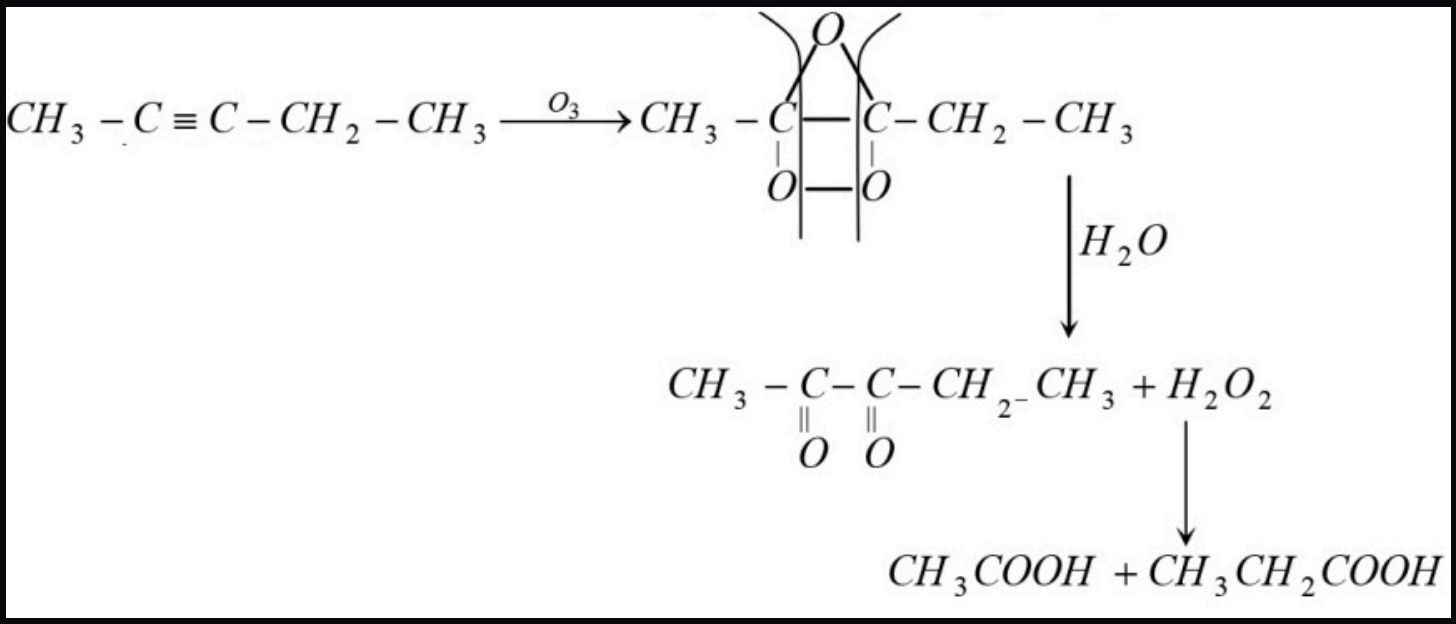
$$= -205 - 152 = -357 \text{ kJ mol}^{-1}$$

$$\Delta H^{\circ}_{hyd} = \frac{-357}{3} = -119 \text{ kJ/ mole}$$

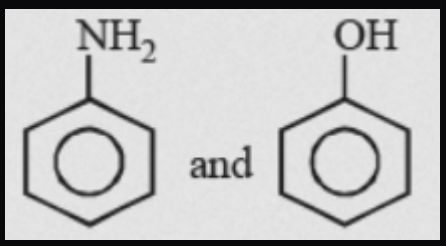
Q.35 Answer:

CH_3COOH, $HO\,OCCH_2CH_3$

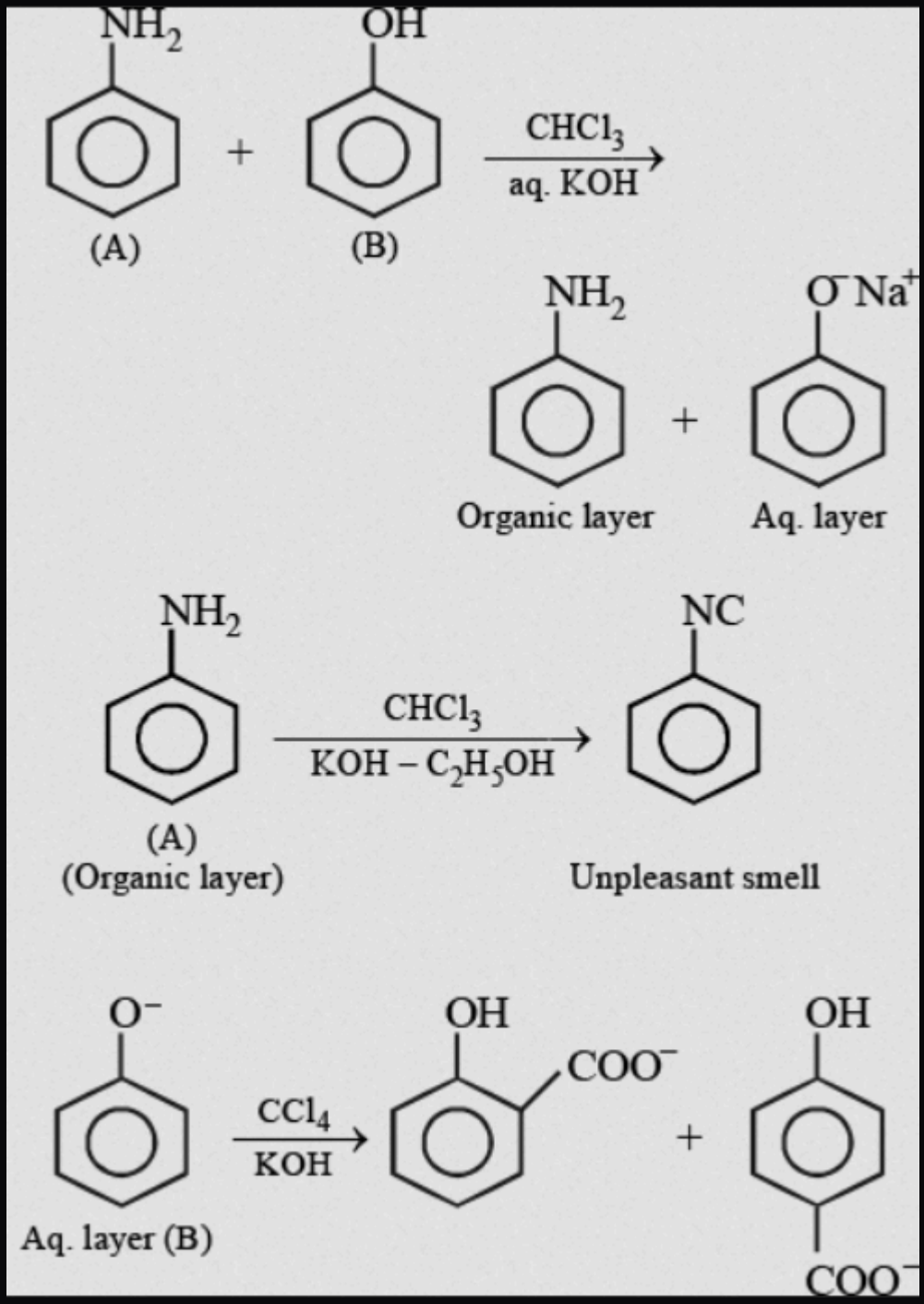
Solution:



Q.36 Answer:



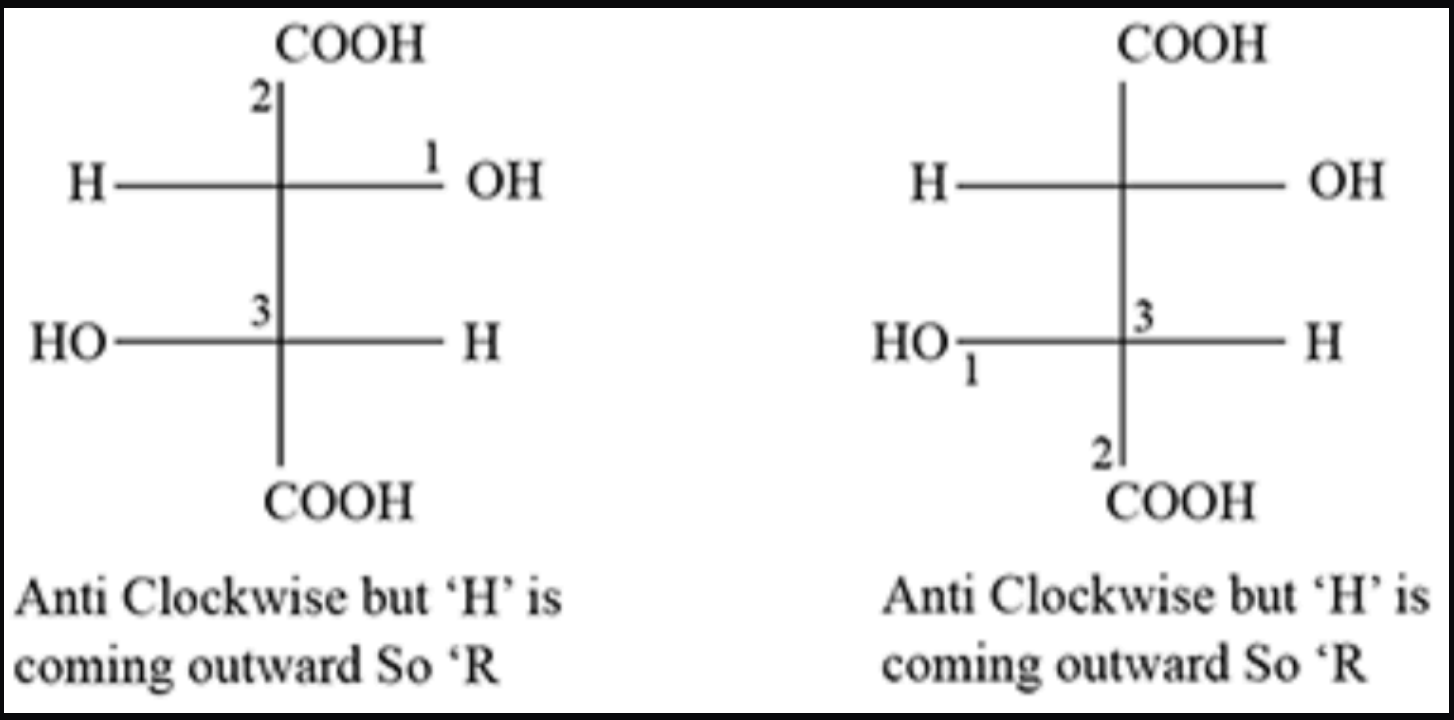
Solution:



Q.37 Answer:

R, R

Solution:



Q.38 Answer:

Statement I is true, Statement II is true; Statement II is not the correct explanation for Statement I.

Solution:

A catalyst provides the reaction with an alternate path that requires a lesser amount of activation energy. Due to this, the rate constant increases and the reaction reaches completion faster. Catalyst does not affect equilibrium constant so yield of product is unchanged.

Q.39 Answer:

Statement - 1 is True, Statement - 2, is False.

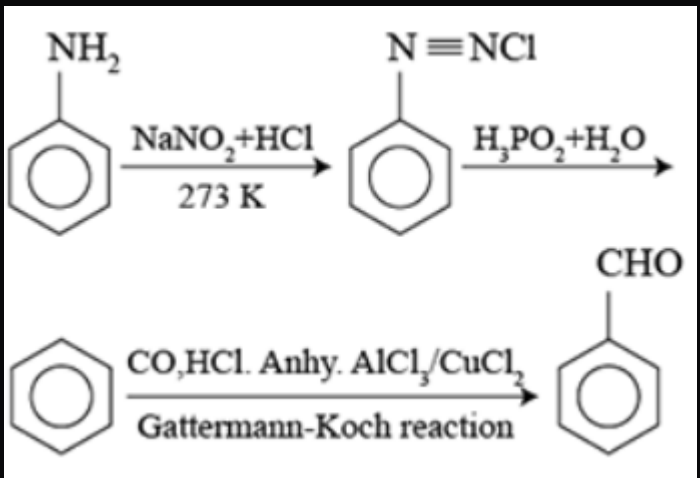
Solution:

Vapour pressure will increase because solute will also contribute in total vapour pressure along with solvent. And there may be less force of attraction between solute and solvent particles.

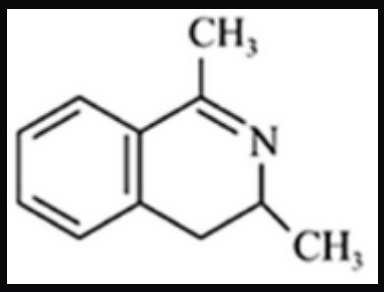
Q.40 Answer:

C_6H_5CHO

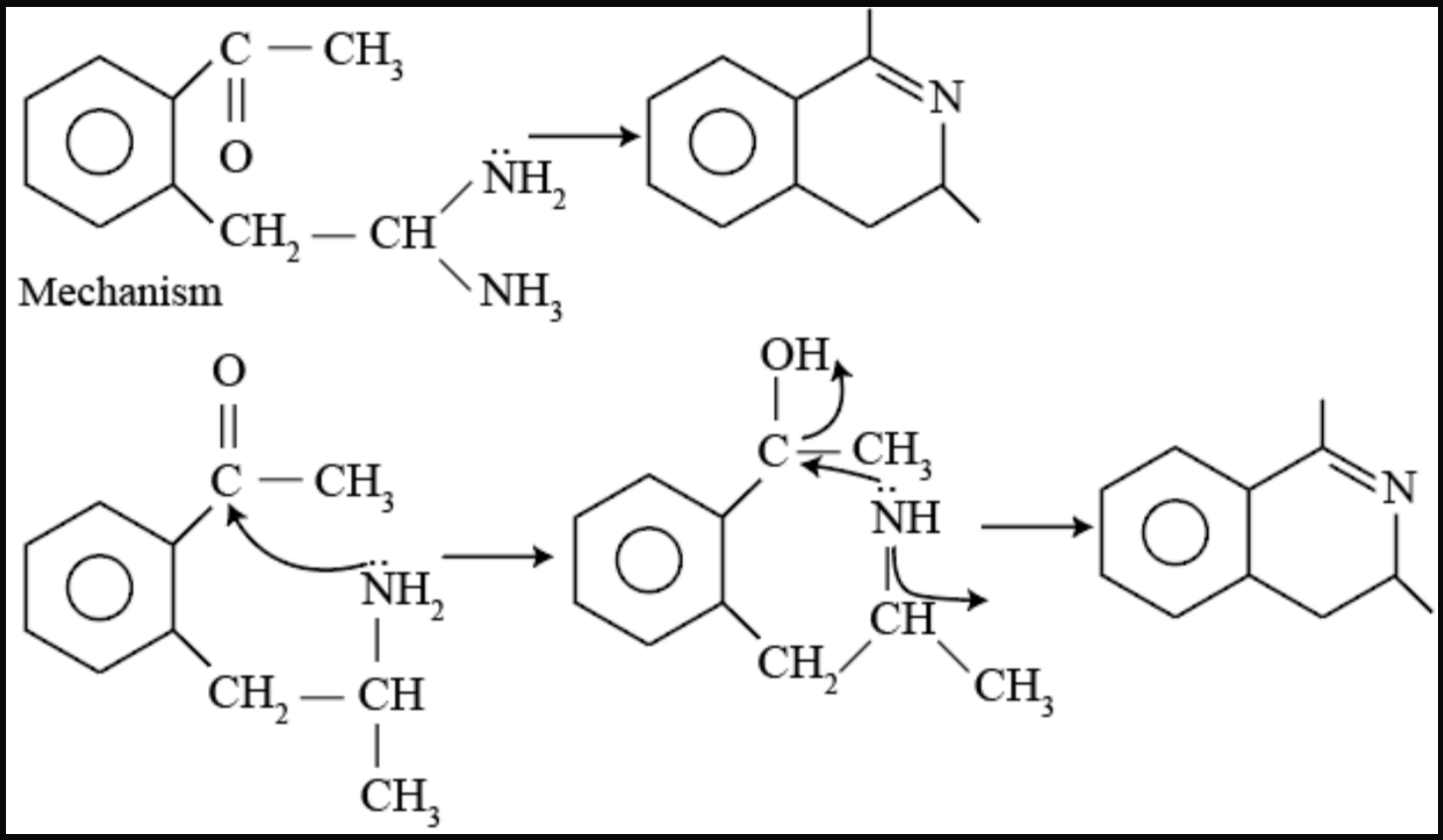
Solution:



Q.41 Answer:



Solution:



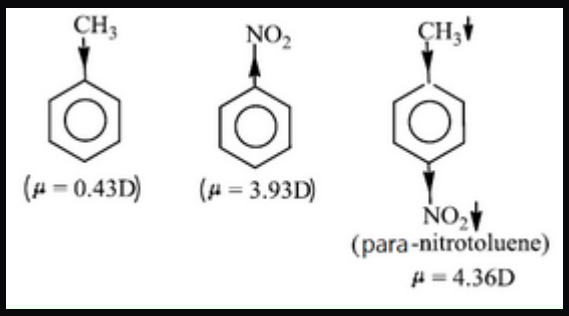
Q.42 Answer:

4.36D

Solution:

Methyl group has $+\mathrm{I}$ effect and $-\mathrm{NO_2}$ group has $-\mathrm{I}$ effect. Therefore, in p-nitro toluene the dipole moments of $-\mathrm{CH_3}$ and $-\mathrm{NO_2}$ groups act in the same direction. So, the resultant dipole moment is additive.

i.e., $3.93 + 0.43 = 4.36$ debye



Q.43 Answer:

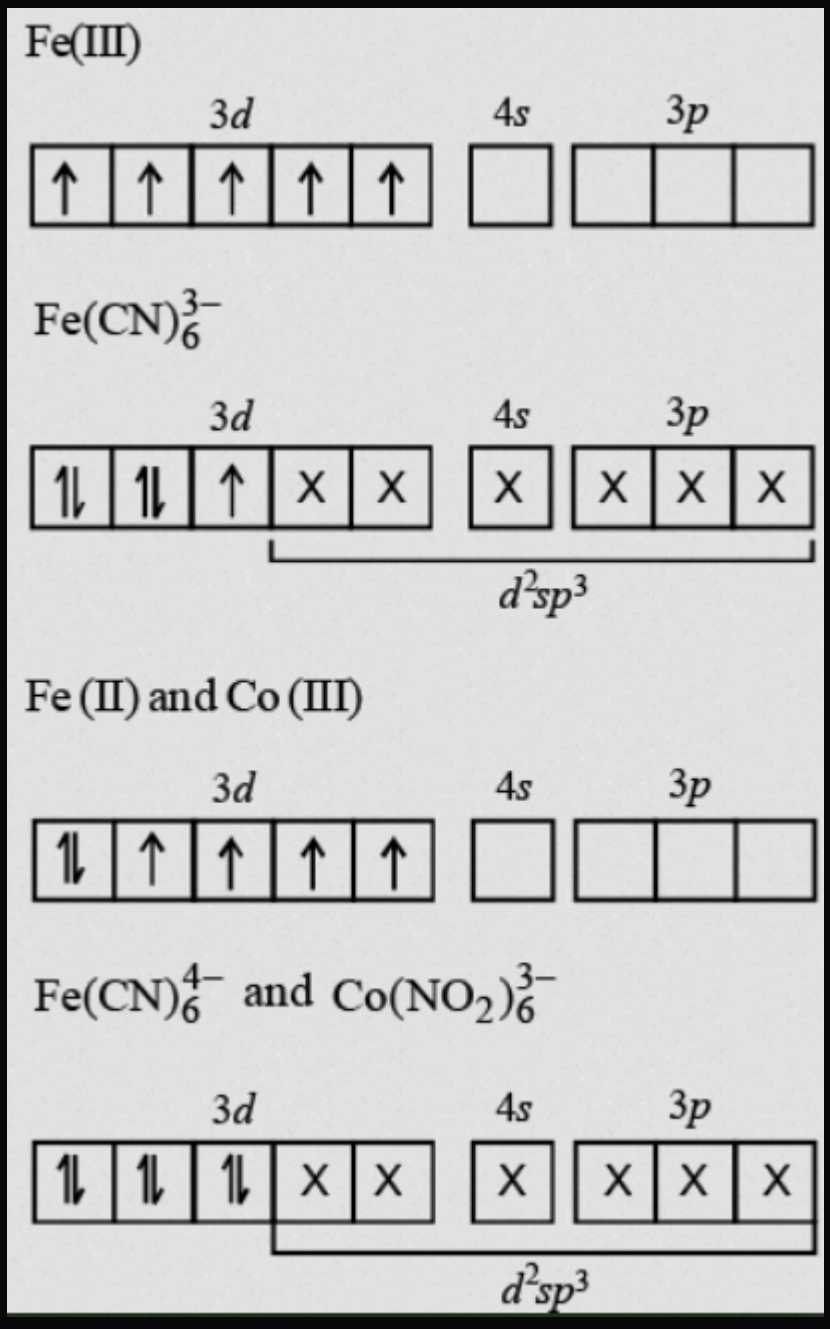
all d^2sp^3

Solution:

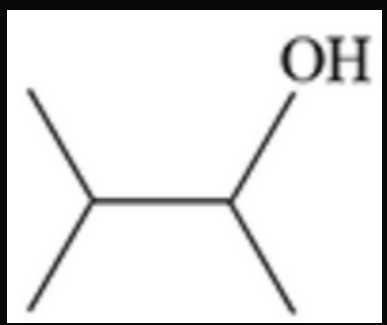
In $Fe(CN)_6^{3-}$ the oxidation number of $Fe = +3$ and so it has a $3d^5$ configuration

In $Fe(CN)_6^{4-}$ the oxidation number of $Fe = +2$ and so it has a $3d^6$ configuration.

In $Co\ (NO_2)_6^{3-}$ the oxidation number of $Co = +3$ and so it has a $3d^6$ configuration.



Q.44 Answer:



Solution:

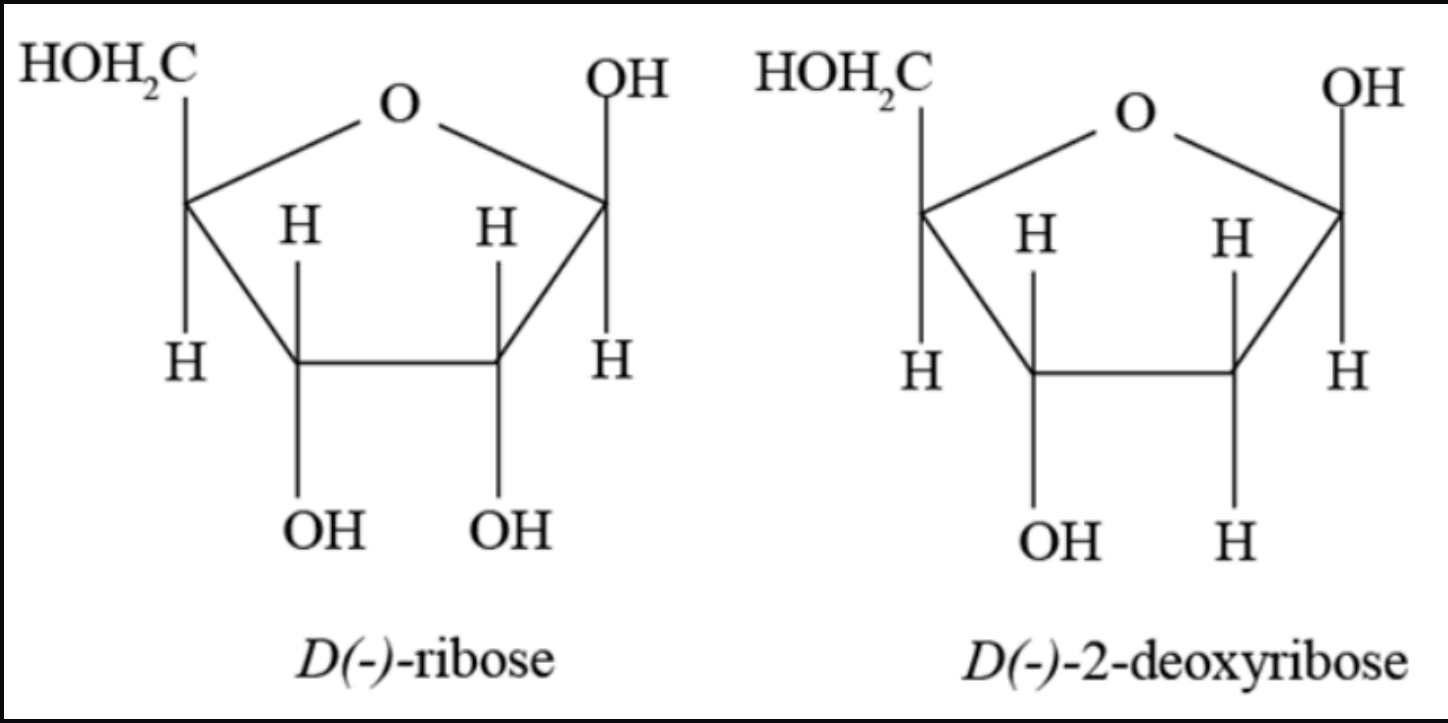
Anti-Markonikov's addition of H and OH

Q.45 Answer:

D-sugar component

Solution:

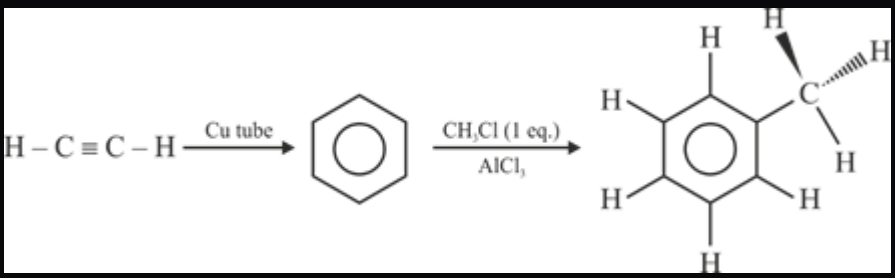
The constituents of nucleic acids are nitrogenous bases, sugar and phosphoric acid. The sugar present in DNA is $D(-)-2$-deoxyribose and the sugar present in RNA is $D(-)$ ribose. Due to these $D(-)$-sugar components, DNA and RNA molecules are chiral molecules.



Q.46 Answer:

—

Solution:



Number of atoms in one plane= 13

Q.47 Answer:

—

Solution:

ΔT_f = freezing point of H_2O - freezing point of ethylene glycol solution $= 0 - \left(-6^{\circ}\right) = 6^{\circ} C$

$K_f = 1.86 \text{ K kg mol}^{-1}$

w_1 = Mass of ethylene glycol in grams

w_2 = Mass of solvent (H_2O) in grams $= 4000 \text{ g}$

m_1 = Molar mass of ethylene glycol $= 62 \text{ g mol}^{-1}$

i = van't Hoff factor $= 1$ ($\because$ ethylene glycol is non-electrolyte)

From, $\Delta T_f = \frac{1000 K_f \text{ W}_1 (i)}{m_1 \text{w}_2}$

$\therefore 6 = \frac{1000 \times 1.86 \times w_1 \times 1}{62 \times 4000}$

$w_1 = 800 \text{ g}$

Q.48 Answer:

—

Solution:

The total number of isomers for the complex compound $\left[Cu^{II}(NH_3)_4\right]\left[Pt^{II}Cl_4\right]$ is four. These four isomers are

$[Cu(NH_3)_3 Cl][Pt(NH_3)Cl_3]$
$[Cu(NH_3)Cl_3][Pt(NH_3)_3 Cl]$
$[CuCl_4][Pt(NH_3)_4]$
and $[Cu(NH_3)_4][PtCl_4]$

The isomer $[Cu(NH_3)_2 Cl_2][Pt(NH_3)_2 Cl_2]$ does not exist due to both parts being neutral.

Q.49 Answer:

—

Solution:

$\Delta E = 13.6 \times 1^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = 12.09 \text{ eV}$

This will be only absorbed, rest is useless (unabsorbed) so, from, $n_2 = 3$ to $n_1 = 1$, we have λ corresponding to $3 \rightarrow 2$; $2 \rightarrow 1$ and $3 \rightarrow 1$. So, total three spectral line.

Q.50 Answer:

—

Solution:

Given $\mathrm{A} = \mathrm{t}_{1/2} = 54\,\mathrm{min}$

$\mathrm{B} = \mathrm{t}_{1/2} = 18\,\mathrm{min}$

$\mathrm{t} = 0\,'\,\mathrm{x}'\mathrm{M} \quad \mathrm{t} = 0'\mathrm{x}\,'\,\mathrm{M}$

$\Rightarrow$ **To calculate :** $[\mathrm{A}_\mathrm{t}] = 16 \times [\mathrm{B}_\mathrm{t}]$ **time** $=$ **?**

$\Rightarrow$ **For** I **order kinetic :** $[\mathrm{A}_\mathrm{t}] = \frac{\mathrm{A}_0}{(2)^\mathrm{n}}$

$\mathrm{n} \rightarrow$ **no. of Half lives**

$\Rightarrow$ **Now from the relation (1)**

$\Rightarrow \frac{\mathrm{x}}{(2)^{\mathrm{n}_1}} = \frac{\mathrm{x}}{(2)^{\mathrm{n}_2}} \times 16 \Rightarrow (2)^{\mathrm{n}_2} = (2)^{\mathrm{n}_1} \times (2)^4$

$\Rightarrow \mathrm{n}_2 = \mathrm{n}_1 + 4 \Rightarrow \frac{\mathrm{t}}{\left(\mathrm{t}_{1/2}\right)_2} = \frac{\mathrm{t}}{\left(\mathrm{t}_{1/2}\right)_1} + 4$

$\Rightarrow \mathrm{t}\left(\frac{1}{18} - \frac{1}{54}\right) = 4 \Rightarrow \mathrm{t} = \frac{4 \times 18 \times 54}{36}$

$\Rightarrow \mathrm{t} = 108\,\mathrm{min}$

Q.51 Answer:

141120

Solution:

There are ${}^7\mathrm{C}_5$ ways of selecting the rings to be worn. If $\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}$ are the number of the rings on the fingers, we need to find the non-negative integers such that $a + b + c + d = 5$.

The numbers of such quadruples is ${}^{5+4-1}\mathrm{C}_{4-1} = {}^8\mathrm{C}_3$. For each set of 5 rings, there are $5!$ arrangements. So, the total number of required arrangements is

${}^7\mathrm{C}_5 \times {}^8\mathrm{C}_3 \times 5! = 141120$

Q.52 Answer:

HP

Solution:

We know that if end points of focal chords are $\left(at^2, 2at\right)$ and $\left(a(t')^2, 2at'\right)$ then

$tt' = -1$

Therefore,

$t_1 t_1' = -1$

$t_2t_2' = -1$

$t_3t_3' = -1$

Now, $t_1,\ t_2$ **and** t_3 **are in** AP**, therefore**

$2t_2 = t_1 + t_3$

$\Rightarrow \frac{-2}{t_2'} = -\frac{1}{t_1'} - \frac{1}{t_3'}$

$\Rightarrow \frac{2}{t_2'} = \frac{1}{t_1'} + \frac{1}{t_3'}$

Therefore, $t_1',\ t_2'$ **and** t_3' **are in** HP**.**

Q.53 Answer:

$A = \ln\left(e^2 - 3\right) - 2$

Solution:

$\phi(x) = Ae^x + x$

$A = \int_0^1 e^t \left\{Ae^t + t\right\} dt$

$= A\frac{e^{2t}}{2} + te^t - e^t\Big)_0^1$

$= \frac{A}{2}\left(e^2 - 1\right) + (e - 0) - (e - 1)$

$A = \frac{A}{2}\left(e^2 - 1\right) + 1$

$A\left(1 - \frac{e^2-1}{2}\right) = 1 \Rightarrow A = \frac{2}{3-e^2}$

$\phi(x) = \frac{2}{3-e^2}e^x + x$

$\phi\left(\ell n\left(e^2 - 3\right)\right) = -2 + \ell n\left(e^2 - 3\right)$

Q.54 Answer:

8

Solution:

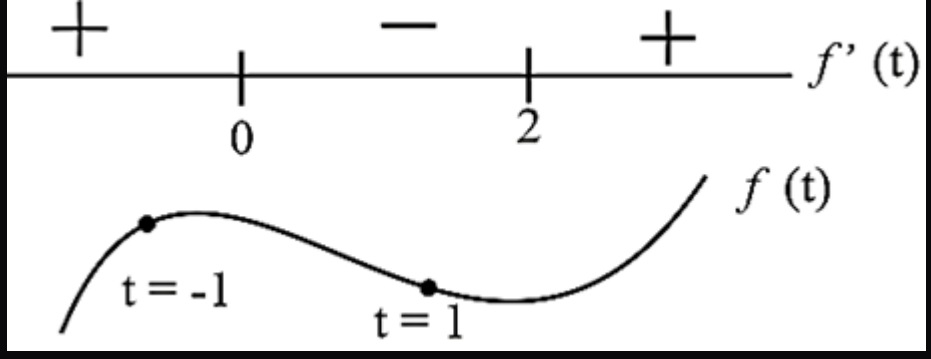
Here, $\text{sgn}(e^x) = 1$

Let $\tan x = t \Rightarrow t \in [-1, 1] \because x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

$\therefore f(t) = 2t^3 - 6t^2 + 2$

$\Rightarrow f'(t) = 2\left[3t^2 - 6t\right]$

$= 6t(t - 2)$



$\therefore f(-1) = 2[-1 - 3 + 1] = -6$

$f(1) = 2[1 - 3 + 1] = -2$

$f(0) = 2$

$\therefore$ **Least value** $= -6$

Local maximum value $= 2$

$\therefore$ **the required positive difference** $= 8$

Q.55 Answer:

2

Solution:

$\tan^{-1}\left(\frac{\left\{\frac{1}{(2x+1)}\right\}+\left\{\frac{1}{(4x+1)}\right\}}{1-\left\{\frac{1}{(2x+1)}\right\}.\left\{\frac{1}{(4x+1)}\right\}}\right) = \tan^{-1}\frac{2}{x^2}$

$\tan^{-1}\left(\frac{6x+2}{8x^2+6x}\right) = \tan^{-1}\frac{2}{x^2}$

Therefore, $\frac{3x+1}{4x^2+3x} = \frac{2}{x^2}$

$x^2(3x+1) = 2\left(4x^2+3x\right)$

$3x^3 - 7x^2 - 6x = 0$

$x\left(3x^2 - 7x - 6\right) = 0$

$x(x-3)(3x+2) = 0$

We take $x = 3, 0$ **because when** $x = -\frac{2}{3}$**, LHS of the given equation will be negative whereas** $\tan^{-1}\left(\frac{2}{x^2}\right)$ **is positive.**

Hence, the number of values $= 2$**.**

Q.56 Answer:

13

Solution:

$(1,1),(2,2),(3,3),(4,4),(2,1),(3,2),(4,3),(1,3),(3,1),(2,4),(4,2)$
$(1,4),(4,1)$

Q.57 Answer:

1

Solution:

Let $\overrightarrow{a}=\lambda^3\hat{i}+\hat{k}$**. Then,**

$\overrightarrow{b}=\hat{i}-\lambda^3\hat{j}$

$\overrightarrow{c}=\hat{i}+(2\lambda-\sin\lambda)\hat{j}-\lambda\hat{k}$

The given vectors are coplanar if

$$\begin{vmatrix}\lambda^3 & 0 & 1\\ 1 & -\lambda^3 & 0\\ 1 & 2\lambda-\sin\lambda & -\lambda\end{vmatrix}=0$$

$\Rightarrow \lambda^3(\lambda^4)+1(2\lambda-\sin\lambda+\lambda^3)=0$

$\Rightarrow \lambda^7+\lambda^3+2\lambda-\sin\lambda=0$

Let

$f(\lambda)=\lambda^7+\lambda^3+2\lambda-\sin\lambda$

As

$\lambda\to\infty,\ f(\lambda)\to\infty$

$\lambda\to-\infty,\ f(\lambda)\to-\infty$

$f(0)=0$

$f'(\lambda)=(7\lambda^6+3\lambda^2+2-\cos\lambda)>0\ \forall\ \lambda\in R\ (\because 7\lambda^6+3\lambda^2+2\geq 2,\ \cos\lambda\in[-1,\ 1])$ **f is strictly increasing on R, hence, the curve meets the x-axis at the origin.** $(\because f(0)=0)$

$\therefore f(\lambda)$ **is equal to zero has only one real solution and that solution is λ equal to zero.**

So, the given vectors are coplanar if λ is equal to zero.

$\therefore$**There exists only one real value of** λ**.**

Q.58 Answer:

9

Solution:

Since, $f(10-x)=f(x)=f(4-x)$

$$\Rightarrow \quad f(10-x)=f(4-x)$$
$$\text{Say,}\quad 4-x=t\Rightarrow f(6+t)=f(t)$$

$\Rightarrow f(x)$ **is periodic function with period 6.**

So, for $x \in [0, 25]$

Total numbers $= 5$

Since, $f(x) = 101$ at $x = 0, 6, 12, 18, 24$

$\Rightarrow f(x)$ is symmetric about $x = 2$ line.

Due to symmetry in one period length. $f(x) = 101$ has one solution at $x = 4$ other than 0 and 6 .

Now, $f(x) = 101$ at $x = 4, 10, 16, 22$

Total numbers $= 4$

Hence, atleast minimum possible number of values of $x = 9$.

Q.59 Answer:

$\frac{5}{8}\sqrt{30}$

Solution:

$|\overrightarrow{AB}| = 3$ and $|\overrightarrow{AC}| = 5$

$\Rightarrow D \equiv \left(\frac{19}{8}, \frac{10}{8}, \frac{17}{8}\right)$

$\left[\because \frac{AB}{AC} = \frac{BD}{DC}\right]$

$|\overrightarrow{AD}| = \frac{5}{8}\sqrt{30} \quad [\text{Let } A \equiv 0]$

Q.60 Answer:

$6, 1$

Solution:

$\therefore$ Let x and y be remaining 2 observation

Mean $= 7 = \frac{2+4+10+12+14+x+y}{7}$

$\Rightarrow x + y = 7$

Variance $= 22 = \frac{1}{7}\left(2^2+4^2+10^2+12^2+14^2+x^2+y^2\right)-(7)^2$

$\Rightarrow x^2+y^2=37$

Now solving (i) and (ii), we get,

$x=1, y=6$ **or** $x=6, y=1$

Q.61 Answer:

$(\sqrt{\mathrm{e}})^{\mathrm{e}^2-2}$

Solution:

In the integral J**, substitute** $\mathrm{x}+1=\mathrm{t}$

$\Rightarrow \mathrm{dx}=\mathrm{dt}$ and $\mathrm{x}^2+2\mathrm{x}=\left(\mathrm{t}^2-1\right)$

Now $J=\int_1^e \frac{e^{\frac{t^2-2}{2}}}{t}dt$ **and** $K=\int_1^e t\ln t e^{\frac{t^2-2}{2}}dt$

Hence $(\mathrm{J}+\mathrm{K})=\int_1^{\mathrm{e}} \mathrm{e}^{\frac{\mathrm{t}^2-2}{2}}\left(\frac{1}{\mathrm{t}}+\mathrm{t}\ln\mathrm{t}\right)\mathrm{dt}$

$=\left(\mathrm{e}^{\frac{\mathrm{t}^2-2}{2}}\ln\mathrm{t}\right)_{\mathrm{t}=1}^{\mathrm{t}=\mathrm{e}}=\mathrm{e}^{\frac{\mathrm{e}^2-2}{2}}=(\sqrt{\mathrm{e}})^{\mathrm{e}^2-2}$

Q.62 Answer:

0

Solution:

Using change of base formula

$\log_b a=\frac{\log b}{\log a}\ ,\ \ a,\ b>0,\ \ a,\ b\neq 1.$

We can write

$$\Delta=\frac{1}{\log x\log y\log z}\begin{vmatrix}\log x & \log y & \log z\\ \log x & \log y & \log z\\ \log x & \log y & \log z\end{vmatrix}=0$$

Q.63 Answer:

27

Solution:

$D=B^T\left(BAB^T\right)\left(BAB^T\right)\left(BAB^T\right)\left(BAB^T\right)\left(BAB^T\right)B$

$D=A^6$

Now, $A^2 = \begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2\sqrt{3}-2 \\ 0 & 1 \end{bmatrix}$

$A^3 = A^2 \cdot A = \begin{bmatrix} 3 & -2\sqrt{3}-2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \sqrt{3} & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3\sqrt{3} & -2\sqrt{3}-8 \\ 0 & 1 \end{bmatrix}$

$A^6 = A^3 \cdot A^3 = \begin{bmatrix} 3\sqrt{3} & -2\sqrt{3}-8 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 3\sqrt{3} & -2\sqrt{3}-8 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 27 & -26-26\sqrt{3} \\ 0 & 1 \end{bmatrix} = D$

$\Rightarrow d_{11} = 27$

Q.64 Answer:

AB and CD are skew-lines and $d_1 = \frac{23}{13}$

Solution:

$\overrightarrow{AB} = 4\hat{i} - 12\hat{j} - 3\hat{k}$

$\overrightarrow{CD} = 3\hat{i} + 4\hat{j} - 12\hat{k}$

$\overrightarrow{AC} = 3\hat{i} - 7\hat{j} + 2\hat{k}$

$\overrightarrow{BD} = 2\hat{i} + 9\hat{j} - 7\hat{k}$

$\therefore d_1 = \dfrac{\left(\overrightarrow{AB}\times\overrightarrow{CD}\right).\overrightarrow{AC}}{\left|\overrightarrow{AB}\times\overrightarrow{CD}\right|} = \dfrac{\left(\overrightarrow{AB}\times\overrightarrow{CD}\right).\overrightarrow{BD}}{\left|\overrightarrow{AB}\times\overrightarrow{CD}\right|}$

$\therefore d_1 = \frac{23}{13}$

Q.65 Answer:

12

Solution:

$|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2$

$= 2\left(|z_1|^2 + |z_2|^2 + |z_3|^2\right) +$

$(z_1\bar{z}_2 + \bar{z}_1 z_2 + z_2\bar{z}_3 + z_3\bar{z}_2 + z_3\bar{z}_1 + z_1\bar{z}_3)$

$= 24 + (z_1\bar{z}_2 + \bar{z}_1 z_2 + z_2\bar{z}_3 + z_3\bar{z}_2 + z_3\bar{z}_1 + z_1\bar{z}_3)$

also $|z_1 + z_2 + z_3|^2 \geq 0$

$\Rightarrow z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_3 + z_3\bar{z}_2 + z_1\bar{z}_3 + z_3\bar{z}_1 \geq -12$

$\therefore |z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2 \geq 12$

Q.66 Answer:

232

Solution:

From the given condition

$^nC_r : {}^nC_{r+1} : {}^nC_{r+2} = 2 : 15 : 70$

$\Rightarrow \frac{^nC_r}{^nC_{r+1}} = \frac{2}{15}$ **and** $\frac{^nC_{r+1}}{^nC_{r+2}} = \frac{15}{70}$

$\Rightarrow \frac{\left(\frac{n!}{(n-r)!.r!}\right)}{\left(\frac{n!}{(n-r-1)!.(r+1)!}\right)} = \frac{2}{15}$ **and** $\frac{\left(\frac{n!}{(n-r-1)!.(r+1)!}\right)}{\left(\frac{n!}{(n-r-2)!.(r+2)!}\right)} = \frac{15}{70}$

$\Rightarrow \frac{(n-r-1)!.(r+1)1}{(n-r)!.r!}\frac{2}{15}$ *and* $\frac{(n-r-2)!.(r+2)!}{(n-r-1)!.(r+1)!} = \frac{3}{14}$

$\Rightarrow \frac{(n-r-1)!.(r+1).r!}{(n-r).(n-r-1)!.r!} = \frac{2}{15}$ **and** $\frac{(n-r-2)!.(r+2).(r+1)!}{(n-r-1).(n-r-2)!.(r+1)!} = \frac{3}{14}$

$\Rightarrow \frac{r+1}{n-r} = \frac{2}{15}$ **and** $\frac{r+2}{n-r-1} = \frac{3}{14}$

$\Rightarrow 17r = 2n - 15$ **and** $17r = 3n - 31$

$\Rightarrow 3n - 31 = 2n - 15, \Rightarrow n = 16$ **and** $r = 1$

Hence, average $= \frac{^nC_r + {}^nC_{r+1} + {}^nC_{r+2}}{3}$

$= \frac{^{16}C_1 + {}^{16}C_2 + {}^{16}C_3}{3}$

$= 232.$

Q.67 Answer:

$\frac{1}{2\sqrt{2}}$

Solution:

The required area is

$A = \int_0^{\frac{\pi}{4}} \left(\tan^3 x + \tan x\right) dx$

$= \int_0^{\frac{\pi}{4}} \tan x \left(\tan^2 x + 1\right) dx$

$= \int_0^{\frac{\pi}{4}} \tan x \cdot \sec^2 x dx$

Let $\tan x = t$

$\Rightarrow \sec^2 x dx = dt$

$\Rightarrow I = \int_0^1 t dt = \left[\frac{t^2}{2}\right]_0^1$

$= \frac{1}{2}$

$\therefore g(x) = \frac{1}{2}\sin x \leq \frac{1}{2 \cdot \sqrt{2}}$

Q.68 Answer:

$(m+1)y = x^{m+1}\cos x + c(m+1)\cos x$

Solution:

This is the linear equation of the form $\frac{dy}{dx} + Py = Q$,

where $P = \tan x$ **and** $Q = x^m \cos x$

Now integrating factor (I.F.) $= e^{\int P dx} = e^{\int \tan dx}$

$= e^{\log \sec x} = \sec x$

Thus sol. is given by, $y.e^{\int P dx} = \int Q \cdot e^{\int P dx} dx + c$

$\Rightarrow y \cdot \sec x = \int x^m \cdot \cos x \cdot \sec x dx + c$

$\Rightarrow y \sec x = \frac{x^{m+1}}{m+1} + c$

$\Rightarrow (m+1)y = x^{m+1}\cos x + c(m+1)\cos x$

Q.69 Answer:

$\frac{47}{87}$

Solution:

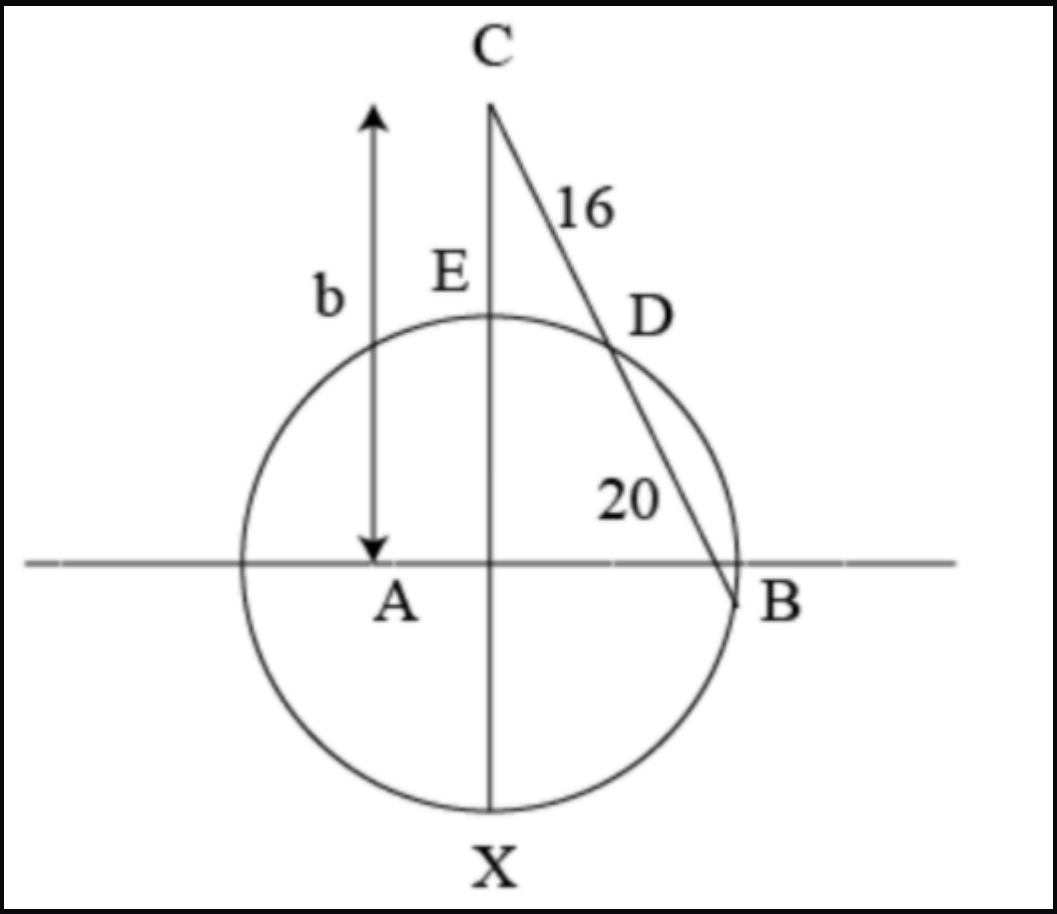
$a^2 - b^2$ **is divisible by '3' if either both a and b are divisible by 3 or neither of them is divisible by '3'**

$n(S) = {}^{30}C_2,\ n(E) = {}^{10}C_2,\ P(E) = \frac{47}{87}$

Q.70 Answer:

$6\sqrt{10}$ units

Solution:



$b^2 + r^2 = (36)^2 \ldots (\text{i})$

Also $CD \cdot CB = CE \cdot CX$

$16 \cdot 36 = (b - r)(b + r)$

$\therefore b^2 - r^2 = 16 \cdot 36$

From (i) and (ii) $2b^2 = 36(36 + 16) = 36 \cdot 52$

$b^2 = 36 \cdot 26 \Rightarrow b = 6\sqrt{26}$ **units**

and $r = \sqrt{(36)^2 - (36)(26)} = 6\sqrt{10}$ **units**

Q.71 Answer:

—

Solution:

Case–I: $x + 2y - 3 = 0$ **&** $2\lambda x + y - 2 = 0$ **are parallel**
$\Rightarrow \frac{2\lambda}{1} = \frac{1}{2} \Rightarrow |\lambda| = 0.25$
Case–II: $3x - y - 1 = 0$ **&** $2\lambda x + y - 2$ **are parallel**
$\Rightarrow \frac{2\lambda}{3} = \frac{1}{-1} \Rightarrow |\lambda| = 1.5$
Case–III: Lines $x + 2y - 3 = 0,\ 3x - y - 1 = 0$ **&** $2\lambda x + y - 2 = 0$ **are concurrent**

$\Rightarrow \begin{vmatrix} 2\lambda & 1 & -2 \\ 1 & 2 & -3 \\ 3 & -1 & -1 \end{vmatrix} = 0 \Rightarrow -10\lambda - 8 + 14 = 0 \Rightarrow |\lambda| = 0.6$

$\Rightarrow 0.25 + 1.5 + 0.6 = 2.35$**=K**

100K = 235.

Q.72 Answer:

—

Solution:

$\sum_{K=1}^{5} {}^{20}C_{2K-1} = 2^{18} \Rightarrow 2^{108} = 2^3 \left(2^5\right)^{21} = 8(33-1)^{21}$

Remainder $= -8$ **or 3**

Q.73 Answer:

—

Solution:

A number of divisible by 3 If its sum of digits are divisible by 3.

So the number can be formed by using the digits.

0, 1, 2, 3 or 0, 2, 3, 4 or 0, 1, 3, 5.

or 1, 2, 4, 5 or 0, 3, 4, 5

Number of 4 digits number divisible by 3

= 4[(4! – 3!)] + 4! = 96 = 6p^2

∴ p = 4 and so p + 4 = 8

Q.74 Answer:

—

Solution:

Let $z = x + iy$

∴ points will be $(x, y), (y, -x), (-x - y, -y + x)$

Area of the triangle for these points as vertices $= \frac{3}{2}\left(x^2 + y^2\right)$ **sq. units** $\because |z| = \sqrt{x^2 + y^2} = 2 \Rightarrow x^2 + y^2 = 4$

$\because |z| = \sqrt{x^2 + y^2} = 2$

$\Rightarrow x^2 + y^2 = 4$

∴ Area of triangle $= 6$**sq. units**

Q.75 Answer:

—

Solution:

Here, $b^2 = ac, 2(b+8) = a+c,$

$(b+8)^2 = a(c+64)$

$\Rightarrow b^2 + 64 + 16b = ac + 64a$

$\Rightarrow b + 4 = 4a \Rightarrow b = 4a - 4$

$2(b+8) = a+c \Rightarrow 2(4a+4) = a+c$

$\Rightarrow c = 7a + 8$

$b^2 = ac \Rightarrow (4a-4)^2 = a(7a+8)$

$\Rightarrow 16a^2 + 16 - 32a = 7a^2 + 8a$

$\Rightarrow 9a^2 - 40a + 16 = 0$

$\Rightarrow (9a-4)(a-4) = 0 \Rightarrow a = \frac{4}{9}, 4$

But for $a = \frac{4}{9} \Rightarrow b < 0$

So, $a = 4 \Rightarrow b = 12$ **and** $c = 36$

$\Rightarrow k = \frac{a+b+c}{3} = \frac{52}{3}$

$\Rightarrow \frac{3k}{13} = \frac{3}{13} \times \frac{52}{3} = 4$